A Joule–Thomson cooler solver

from a handwritten derivation to a validated design tool

A development journal

How a reading of Zhang & Dang's 1 K hybrid cryocooler paper (*Cryogenics* **116**, 2021) turned into a general solver for precooled JT coolers: what was thought, discussed, built, broken, fixed and checked, and why each choice was made.

Written for my own future reference. Part I is the story; Part II is the technical reference.

Started as the ME 623 (Cryogenics) paper study, Semester 5. Compiled October 2026.

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Before you read

This is the record of a project that started as a seminar on one paper and ended as a small piece of software: a solver for Joule–Thomson (JT) coolers that are precooled by a multi-stage cryocooler. The seminar asked for an explanation of the paper. Doing it properly meant re-deriving the paper, then building something that could reproduce it, then checking that the same thing worked on other people's coolers, and finally using it to answer design questions the paper leaves open.

The results are in Part II and in the project folder. What is easy to lose is *how* we got there: what the first idea was, what was argued about, what was tried and thrown away, which bugs taught something, and why the code is shaped the way it is. That is what Part I is for. If you come back to this in five years, read Part I first, with the paper open beside you.

Where the material comes from

The story is reconstructed from the following, all kept in the folder `dev_history/`:

Source	What it contains
`handwritten_derivation.pdf`	Eight pages written while first studying the paper: objective, assumptions, the CHEX, the cold-end analysis, the optimum, the nozzle. Reproduced in Chapter 2.
`gpt_chat_architecture_and_cases.txt`	The first coding conversation: the 14 states, the CHEX as a block, forward and inverse cases, generalisation to n stages.
`gpt_chat_reproduction_and_property_request.txt`	The same conversation later: the reproduction script, the decision to use HEM, the He-3 property request spreadsheet.
`claude_planning_chat.txt`	A parallel planning conversation: layered pseudo-code, corrections, validation targets, open decisions, the first He-3 interpolator.
`claude_this_session_start_and_end.txt`	The start and end of the long working session in which the code was reviewed, fixed, connected to real He-3 data, validated, extended and documented.
`terminal_output_user_run.txt`	My own run of the whole validation chain on my laptop, which matched the numbers produced during development.

Where the record of a decision does not include its reason, no reason is given here. Numbers quoted in Part I are the ones the code produces; they are repeated with full context in Part II.

How the boxes are used

Insight

Something understood along the way that is worth remembering on its own.

Decision

A choice that shaped the code, with the reason.

What went wrong

A mistake, a bug or a wrong turn, and what it taught.

In the conversation

Words as they were said or written at the time.

Part I

The story

Chapter 1

The paper and the question

1.1 Why a 1 K hybrid cooler at all

Zhang & Dang start from a practical problem. Detectors such as superconducting nanowire single-photon detectors want temperatures around 1–2 K. Liquid-helium baths with pumped vapour reach this, but they evaporate, need refilling and are impossible in orbit. Two-stage Gifford–McMahon coolers reach about 2.1 K without load, slightly too warm once a real load is applied, and they vibrate and wear. Stirling-type pulse-tube coolers (SPTC) have no moving cold parts and live long, which suits space, but even four-stage SPTCs stop at about 3 K (Nast et al., 2008; Dang et al. 3.3 K in 2020): the regenerator loses effectiveness at these temperatures.

A JT cooler does not have this problem. It is *recuperative*: a counterflow heat exchanger, not a regenerator, and the cold is produced by isenthalpic expansion through a small restriction. Below 2 K this works, but only if the gas arriving at the restriction is already cold. So the natural machine is a *hybrid*: the SPTC does the work from 300 K down to a few kelvin, and a JT loop does the last step. In the paper's design the last three cold heads of a four-stage SPTC are the three precoolers (PHEX) of a He-3 JT loop with four counterflow exchangers (CHEX), a JT valve and an evaporator at 1 K.

Why He-3? At 1 K its saturation pressure is 1.16 kPa, which is a pressure a return line and a compressor can actually handle; He-4 at 1 K would need far lower pressure. The paper argues (with a 0.35 kPa pressure drop reported by Jahromi and Miller) that the return-line pressure drop can be kept small enough to neglect it in the analysis. Keep this in mind: it is the assumption that later decides how well we can predict real, measured coolers (Chapter 11).

1.2 What the paper does

The paper is Part A (theory and modelling) of a pair. Its argument has five steps, and our work follows them one by one:

1. **Enthalpy flow model** (Sect. 3.1, Eqs. 1–22). Write the cycle as 14 states on the P–h plane, define the efficiency of each CHEX, and express the refrigeration q_C and the precooling loads $q_{\mathrm{pre},i}$ through enthalpies. In the ideal limit the loads and the refrigeration are tied by a global balance (Eq. 22).
2. **Optimum of the last stage** (Sect. 3.2, Eqs. 24–36). Only the last CHEX, the valve and the evaporator matter for q_C. With an ideal last CHEX $q_C = \min(\Delta h_T, q_L)$; the best P_U for a given precooling temperature is where $\Delta h_T = q_L$. For $T_C = 1$ K and $T_{\mathrm{pre3}} = 8$ K: $P_U = 0.36$ MPa, $q_C = 8.0$ kJ/kg. Fits: Eq. 35 ($q_{C,\mathrm{opt}}$), Eq. 36 ($P_{U,\mathrm{opt}}$ as a quartic in T_{pre3}).
3. **Mass flow** (Sect. 4, Eqs. 41–49). The JT valve as a choked nozzle; real-gas flux through a deviation factor Γ on top of the ideal-gas flux; fitted as $G = \alpha P_U^{\beta} + \gamma$ (Eq. 49, Table 2) for $T_C = 1, 2, 2.6, 3$ K.
4. **Gross cooling capacity** (Sect. 5). $Q_C = \dot{m} q_C$. With $d = 20$ µm: $\dot{m} = 2.34$ mg/s and $Q_C = 18.7$ mW at the optimum.
5. **Imperfect CHEX** (Sect. 6). With $\eta_4 = 0.97$, Q_C falls to 14.7 mW (about 21 %); the cold- and warm-end temperature differences of the last CHEX should not exceed 1 K and 0.2 K.

1.3 The question we asked

Reading the paper raises a few uncomfortable questions. The optimum is found for the last stage only; what about the other three CHEXs and the two warmer precoolers, whose temperatures the paper never gives? The mass flux uses a method that needs a sound speed for He-3 that we did not have. The numbers come from He-3 properties that are not in any standard package. And would the same reasoning work for the He-4 coolers the paper cites?

So the goal became, in the words of the first handwritten page:

Handwritten, page 1

Objective will be to find P_U, T_i, $Q_{\text{pre},i}$ such that the Stirling pulse tube & JTC could be fully designed, just based on T_C & Q_C required.

And later, in the first coding conversation:

In the conversation

I see my goal as modelling the whole JT, all the 14 stages. I see inputs as $T_{\text{pre1,2,3}}$ and P_U, and T_C, and $\eta_{\text{chex1,2,3,4}}$ for a target T_C and Q_C.

That is the project in two sentences: a model of the *whole* JT loop, from which the demands on the precooling cryocooler can be read off.

Chapter 2

The handwritten derivation

Everything started on eight sheets of paper. They are reproduced here, page by page, with what each step says, what it meant at the time, and what it turned into later. The derivation is the paper's, re-done by hand until it made sense; the order and the emphasis are mine.

2.1 Page 1: objective, schematic, hard assumptions

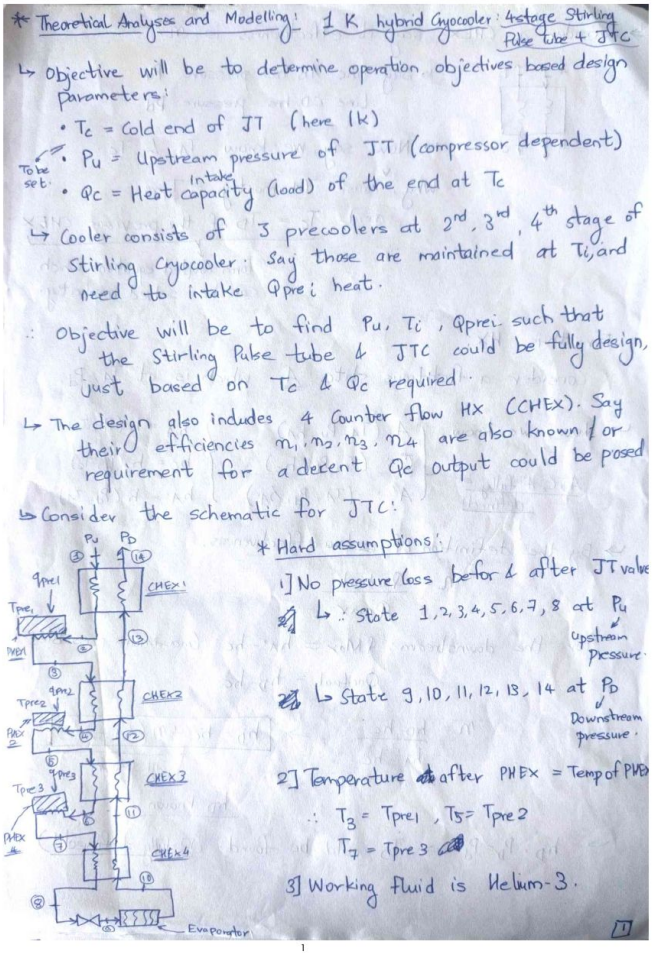


Figure 2.1: Handwritten page 1.

The page fixes the vocabulary that survived all the way to the code. T_C is the cold end of the JT (1 K here). P_U is the upstream pressure, "compressor dependent", i.e. to be set. Q_C is the heat intake at T_C. The cooler has three precoolers at the 2nd, 3rd and 4th stages of the Stirling cooler, held at T_i and absorbing $Q_{\text{pre},i}$. There are four CHEXs with efficiencies $\eta_1 \dots \eta_4$, "known, or a requirement for a decent Q_C could be posed".

The schematic numbers the states exactly as the paper does: 1 (top, P_U) down to 8 before the valve, 9 after it, 10 leaving the evaporator, 11–14 on the way back up. The "hard assumptions" are three:

1. no pressure loss before and after the JT valve, so states 1–8 are at P_U and 9–14 at P_D;
2. the temperature after a PHEX is the PHEX temperature: $T_3 = T_{\text{pre1}}$, $T_5 = T_{\text{pre2}}$, $T_7 = T_{\text{pre3}}$;
3. the working fluid is He-3.

Insight: what assumption 2 really says

Assumption 2 is what makes the problem tractable. The precooler is not modelled; it is a boundary condition that *pins* the temperature of the supply stream at three places. Everything else then follows from energy balances. It also means that the precooler loads are *outputs*: whatever heat has to be removed to bring the gas to $T_{\text{pre},i}$ is what the cryocooler stage must absorb. That is the link from the JT model to the cryocooler, and the reason the final tool produces a "load map".

2.2 Page 2: one counterflow exchanger

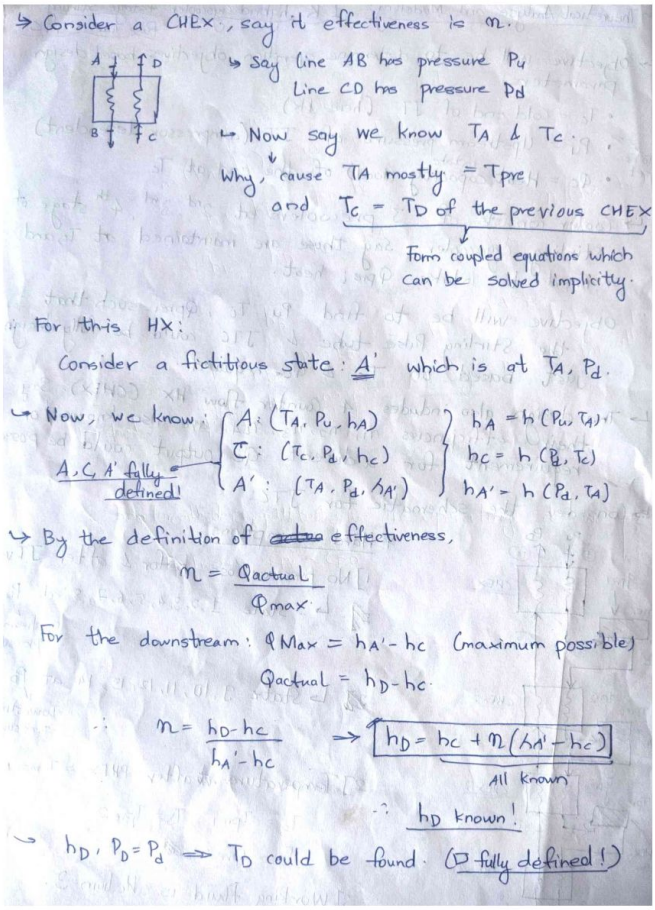


Figure 2.2: Handwritten page 2.

A CHEX is drawn as a box with four ends: A hot in and B hot out on the P_U line, C cold in and D cold out on the P_D line. "Say we know T_A and T_C. Why? Because T_A is mostly T_{pre} and $T_C = T_D$ of the previous CHEX" — the inlets of each exchanger are known, either from a precooler or from the neighbouring exchanger. The page notes these "form coupled equations which can be solved implicitly".

Then the key construction: a fictitious state A' at (T_A, P_D), the cold stream brought to the hot inlet temperature. The most the cold stream could gain is $h_{A'} - h_C$, so with the effectiveness defined as actual over maximum heat,

$$\eta = \frac{Q_{\text{actual}}}{Q_{\max}} = \frac{h_D - h_C}{h_{A'} - h_C} \quad \Longrightarrow \quad h_D = h_C + \eta\,(h_{A'} - h_C),$$

and D is fully defined by h_D and P_D.

Insight: the one assumption hidden in A'

$h_{A'} - h_C$ is the maximum *on the cold side only*: it is the heat needed to warm the return gas up to the supply temperature. It silently assumes that the warm end is where the exchanger runs out of temperature difference. That is true for most of the cooler, but not for every operating point of the last CHEX. Three chapters later this turned out to be the source of a real bug (Chapter 6).

2.3 Page 3: the energy balance and the two pinches

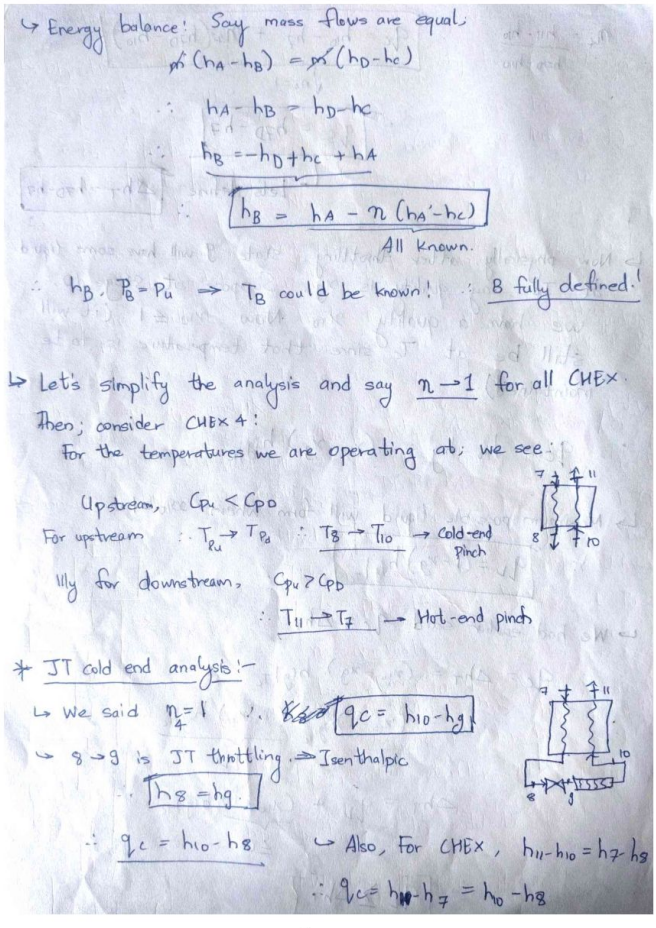


Figure 2.3: Handwritten page 3.

With equal mass flows, $h_A - h_B = h_D - h_C$, so $h_B = h_A - \eta(h_{A'} - h_C)$: all explicit, nothing to iterate. Then "let's simplify the analysis and say $\eta \to 1$ for all CHEX". For CHEX 4 at the operating temperatures, the upstream (high-pressure) stream has the smaller heat capacity, $c_{p,U} < c_{p,D}$, so

$T_8 \to T_{10}$: a *cold-end pinch*. The page also writes the opposite case, $c_{p,U} > c_{p,D}$, giving $T_{11} \to T_7$, a *hot-end pinch*.

The JT cold-end analysis follows: with $\eta_4 = 1$, $8 \to 9$ is throttling, isenthalpic, so $h_8 = h_9$ and $q_C = h_{10} - h_8$; and since $h_{11} - h_{10} = h_7 - h_8$ for the CHEX, $q_C = h_{11} - h_7$.

Insight: which end pinches is decided by the heat capacities

The c_p argument on this page is the physical intuition behind the "two limits" of the final solver. Where the stream with the smaller heat-capacity rate runs out of temperature difference, the exchanger pinches. For He-3, c_p at 0.36 MPa collapses below about 4 K while c_p at 1.16 kPa stays near 7 kJ/kg/K (Fig. 2.4), so the last CHEX tends to pinch at the cold end and the warmer CHEXs at the warm end. The code does not compute c_p at all: it compares the two *enthalpy* limits directly, which is the same argument done exactly instead of with an average c_p.

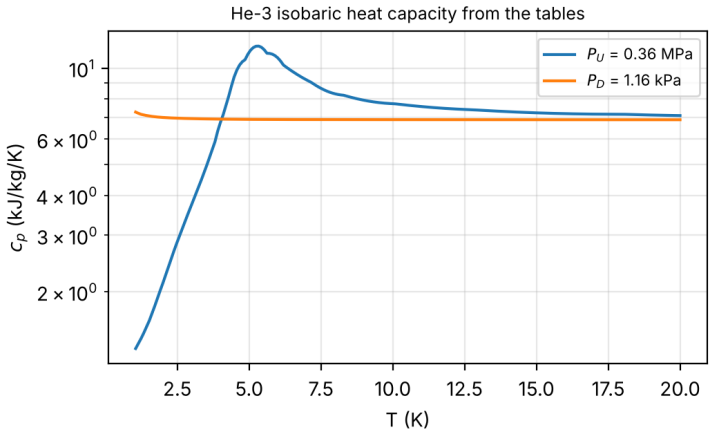


Figure 2.4: He-3 c_p on the two isobars of the paper's cooler, from the property tables. The crossing near 4 K is the handwritten pinch argument made visible.

2.4 Page 4: q_C, Δh_T and the quality argument

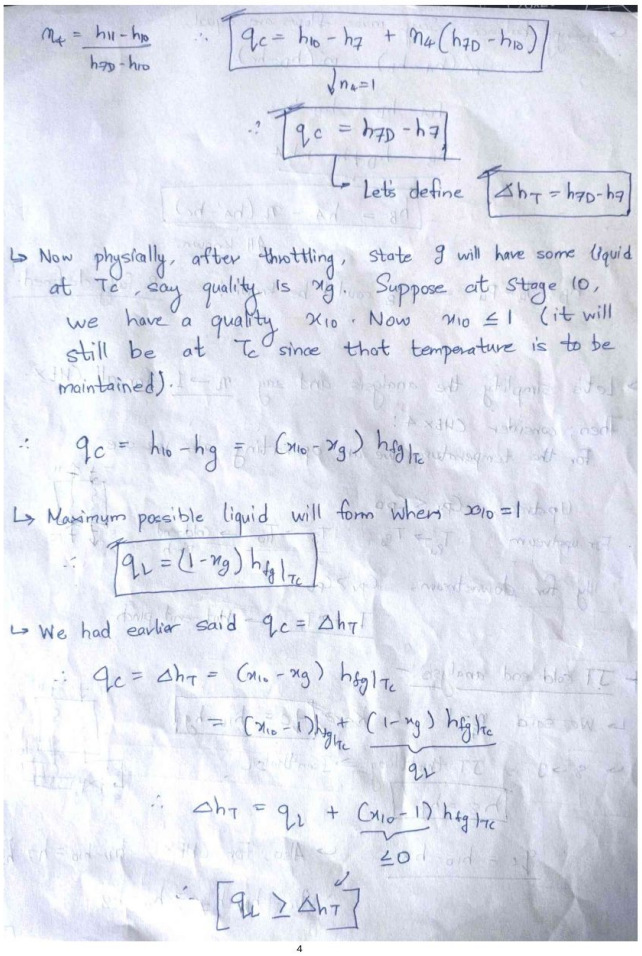


Figure 2.5: Handwritten page 4.

With the paper's definition $\eta_4 = (h_{11} - h_{10})/(h_{7D} - h_{10})$, the specific refrigeration is

$$q_C = h_{10} - h_7 + \eta_4 (h_{7D} - h_{10}) \xrightarrow{\eta_4 = 1} q_C = h_{7D} - h_7 \equiv \Delta h_T.$$

Δh_T is the *isothermal* enthalpy difference between the two pressures at the precooling temperature.

Then a physical check. After throttling, state 9 is two-phase with quality x_9; leaving the evaporator, state 10 has quality $x_{10} \leq 1$ (it is still at T_C). So $q_C = (x_{10} - x_9) h_{fg}$. The most liquid that can be evaporated corresponds to $x_{10} = 1$, giving the *latent limit*

$$q_L = (1 - x_9)\, h_{fg}\big|_{T_C}.$$

Writing $\Delta h_T = q_L + (x_{10} - 1) h_{fg}$ with $(x_{10} - 1) \leq 0$ shows that a physical cycle needs $q_L \geq \Delta h_T$.

2.5 Page 5: why $q_C = \min(\Delta h_T, q_L)$

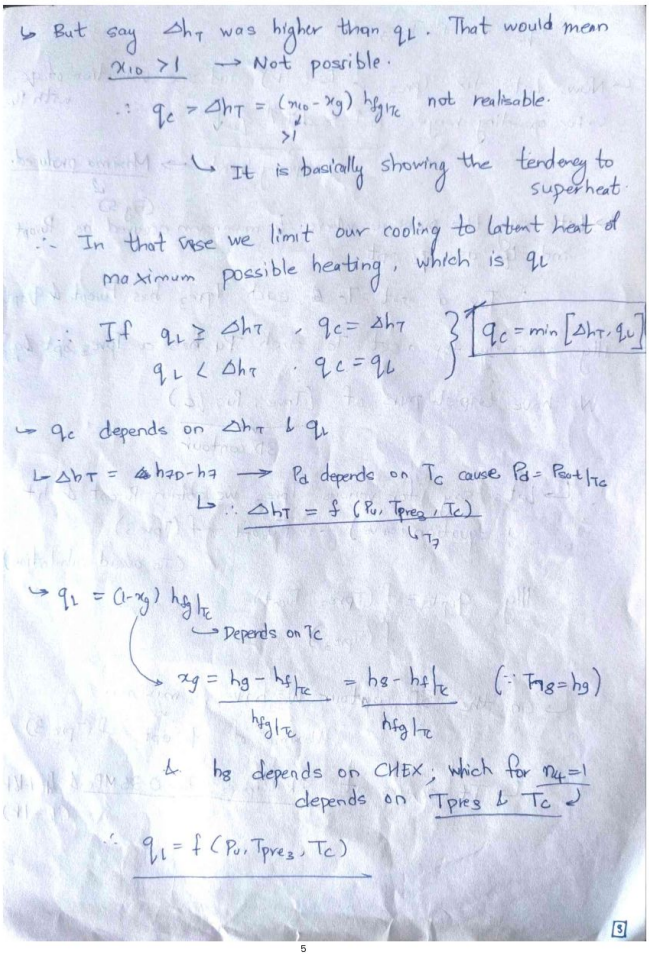


Figure 2.6: Handwritten page 5.

"But say Δh_T was higher than q_L. That would mean $x_{10} > 1$ — not possible." The formula would then claim more refrigeration than there is liquid to evaporate; "it is basically showing the tendency to superheat". So the cooling is capped at the latent limit:

$$q_C = \begin{cases} \Delta h_T, & q_L \geq \Delta h_T \\ q_L, & q_L < \Delta h_T \end{cases} \qquad \Longrightarrow \qquad q_C = \min[\Delta h_T,\, q_L].$$

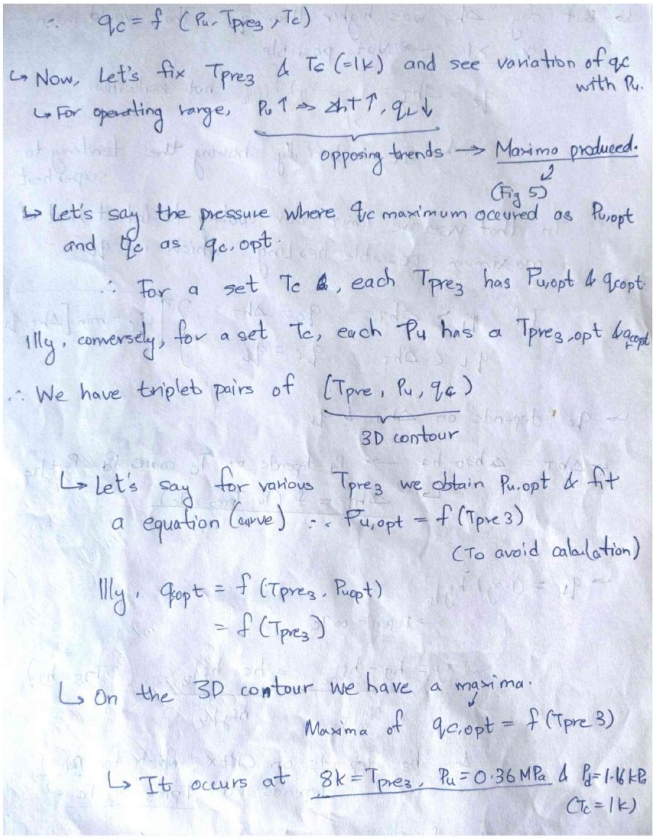
Δh_T depends on P_U, T_{pre3} and T_C (through $P_D = P_{\text{sat}}(T_C)$). q_L depends on $x_9 = (h_8 - h_f)/h_{fg}$, hence on h_8, hence on the CHEX.

Insight: the same rule seen from two sides

On paper, $q_C = \min(\Delta h_T, q_L)$ came from the quality argument: you cannot evaporate more than all the liquid. In the solver it comes out of the exchanger: when $\Delta h_T > q_L$, the formula of page 4 would need the gas leaving the last CHEX to be colder than the evaporator, which no counterflow exchanger can do. Both are true. The quality argument tells you *what* is capped; the exchanger argument tells you *why the formula of page 4 stops being valid*, and that is the version a program needs, because a program will happily evaluate a formula outside its range (Chapter 6).

Note that $(1 - x_9)h_{fg} = h_g - h_8$ exactly, even if h_8 is below the saturated-liquid enthalpy (a subcooled liquid gives $x_9 < 0$). In the code q_L is written $h_g(T_C) - h(P_U, T_C)$, which is the same quantity evaluated where it matters: with the last CHEX pinched at the cold end, $T_8 = T_C$.

2.6 Page 6: the optimum and the fits



6

Figure 2.7: Handwritten page 6.

Fix T_{pre3} and T_C and vary P_U: "$P_U \uparrow \Rightarrow \Delta h_T \uparrow$, $q_L \downarrow$, opposing trends $\Rightarrow$ maxima produced" (the paper's Fig. 5). For each T_{pre3} there is a $P_{U,\text{opt}}$ and $q_{C,\text{opt}}$; conversely for each P_U there is a $T_{\text{pre3,opt}}$. "We have triplet pairs of $(T_{\text{pre}}, P_U, q_C)$: a 3-D contour." Fitting $P_{U,\text{opt}} = f(T_{\text{pre3}})$ and $q_{\text{opt}} = f(T_{\text{pre3}}, P_{U,\text{opt}})$ is "to avoid calculation". "On the 3-D contour we have a maxima … it occurs at 8 K $= T_{\text{pre3}}$, $P_U = 0.36$ MPa, $P_D = 1.16$ kPa ($T_C = 1$ K)."

Insight: a peak or the edge of a plateau?

This is the one place where the solver later changed the *interpretation* of the handwritten notes without changing a number. At $P_U = 0.358$ MPa, q_C is the same 7.98 kJ/kg for every $T_{\text{pre3}} \leq 8$ K and falls above it: 8 K is not a peak but the *warmest* point of a plateau. Colder precooling buys nothing and costs a lot. So "the optimum T_{pre3}" really means "the warmest precooling that still gives the full latent limit". This is why the code calls it $T_{\text{pre,opt}}$ and why, when both P_U and T_{pre} are free, the meaningful objective becomes Q_C, not q_C (Chapter 9).

2.7 Pages 7–8: the valve as a nozzle

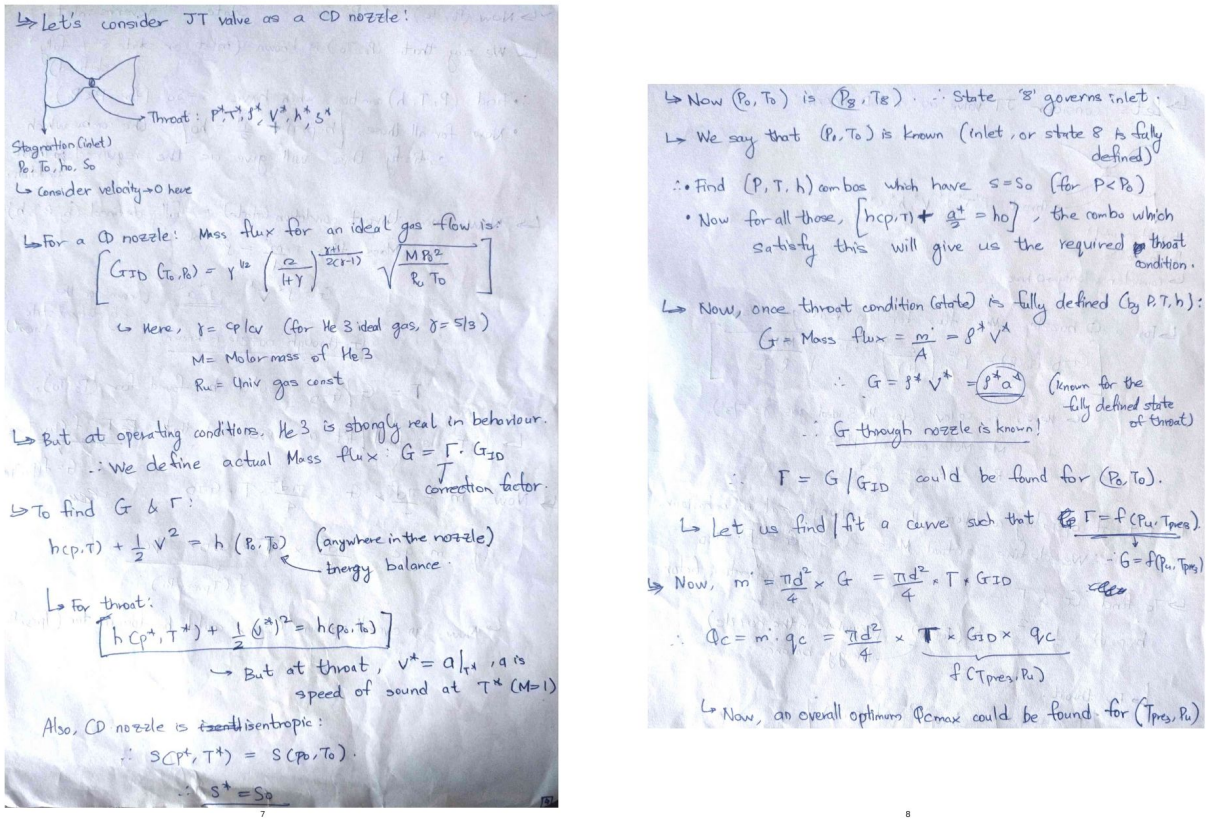


Figure 2.8: Handwritten pages 7 and 8.

The JT valve is treated as a converging–diverging nozzle with stagnation state (P_0, T_0, h_0, s_0) and a throat $(P^*, T^*, \rho^*, V^*, h^*, s^*)$. For an ideal gas the choked flux is

$$G_{ID} = \gamma^{1/2}\Big(\frac{2}{1+\gamma}\Big)^{\frac{\gamma+1}{2(\gamma-1)}}\sqrt{\frac{MP_0^2}{R_u T_0}}, \qquad \gamma = 5/3 \text{ for He-3.}$$

He-3 here is strongly non-ideal, so $G = \Gamma\, G_{ID}$ with a correction Γ. To find it: energy, $h(P,T) + V^2/2 = h_0$; at the throat $V^* = a$ (Mach 1); isentropic, $s^* = s_0$. Procedure: the inlet is state 8, $(P_0, T_0) = (P_8, T_8)$; find all (P, T, h) with $s = s_0$ for $P < P_0$; the one with $h + a^2/2 = h_0$ is the throat; then $G = \rho^* a^*$ and $\Gamma = G/G_{ID}$. Fit $\Gamma = f(P_U, T_{\text{pre3}})$, so $Q_C = \frac{\pi d^2}{4}\,\Gamma\, G_{ID}\, q_C$, and an overall optimum $Q_{C,\max}$ can be found over (T_{pre3}, P_U).

Insight: what the solver kept and what it changed

The solver keeps the physics of these two pages — isentropic expansion from state 8, energy conservation, choking — and drops the machinery. Along the isentrope, the mass flux $G(P) = \rho\sqrt{2(h_0 - h)}$ has a maximum, and $\mathrm{d}G/\mathrm{d}P = 0$ is exactly the condition $V = a$. So instead of computing a sound speed and solving for the throat, the code *maximises* $G(P)$ on a pressure grid. No speed of sound (which we did not have for He-3), no G_{ID}, no Γ, no fit. This is the "HEM" of the later chapters. It reproduces the paper's Table 2 to better than 1 %.

2.8 What the eight pages amount to

Looking back, the handwritten derivation already contained the whole skeleton of the solver: explicit CHEX equations from inlet states, the cold-end analysis, $q_C = \min(\Delta h_T, q_L)$, the optimum as the crossing of two opposing trends, and the choked-flow route to Q_C. What it did not yet have was (i) the realisation that the CHEX definition must respect both ends, (ii) a way to get He-3 properties, (iii) a way to handle the upstream stages and the precooler temperatures the paper leaves blank, and (iv) any check that the result was right. The rest of this journal is about those four things.

Chapter 3

From derivation to architecture

The first coding conversation began with a deliberate restraint: "Just understand the paper layout. We are going to code. I will tell you specifics." The specifics then came one idea at a time, and each became a piece of the architecture.

3.1 The reduced model and the full model

The paper's analysis lives in states 7–11: the last CHEX, the valve and the evaporator. T_C fixes state 10 (saturated vapour at $P_{\text{sat}}(T_C)$), P_U and T_{pre3} fix state 7, and the optimum follows from $\Delta h_T = q_L$. That is enough to reproduce the paper's Figs. 5–8. But the goal was the whole cycle:

In the conversation

Now understand, say PHEX 3, which is between JT state points 6 and 7. Now I can find Q_{pre3} by doing $h_6 - h_7$. Similarly for others. So if I get to model all state points of the JT I can model the cryocooler, and vice versa if I have a cooler, I can counter-model the JT side.

Decision: model all 14 states, not only 7–11

The precooler loads $Q_{\text{pre},3} = \dot{m}(h_6 - h_7)$, $Q_{\text{pre},2} = \dot{m}(h_4 - h_5)$, $Q_{\text{pre},1} = \dot{m}(h_2 - h_3)$ are what connect the JT loop to the cryocooler. They need every state, so the solver must solve every CHEX.

3.2 The CHEX as a reusable block

Next, one exchanger with ends A, B, C, D and negligible pressure drop. Two ways of using it were identified: *known A and C* (both inlets) gives B and D; *known A and B* (the hot side) gives C and D. Then the observation that turns the cycle into a simple chain:

In the conversation

By P_U and T_{pre3} we get state 7 defined and by T_C we have state 10 (quality $x = 1$) at $P_{\text{sat}}|_{T_C}$. It looks exactly as A, C known, η_{chex} known, so finding 8 and 11 is possible. Now consider CHEX 3: we know point 5 since P_U and T_{pre2}, and we know 11, same problem again, and so on for all CHEX.

Insight: why the cycle is solved from the cold end

The evaporator outlet is the one state fixed without knowing anything else. Every CHEX has its hot inlet fixed by a precooler (or by 300 K), and its cold inlet is the cold outlet of the next-colder CHEX. So start at the evaporator and walk up: CHEX 4, then 3, 2, 1. Each step uses only known inlets. This is the loop in `solve_cycle` to this day.

3.3 Two directions: forward and inverse

Two "cases" were then defined.

- **Case 1, forward.** Given T_{pre}, P_U, T_C and η: solve every state, then Q_C and the loads the Stirling cooler must provide.
- **Case 2, inverse.** Given the Stirling cooler's capability (temperature and load per stage): from Q_{pre1}, $h_2 - h_3$ is known; with T_{pre1} and P_U, state 3 and then state 2 are known; so each CHEX has A and B known and C, D are found.

The architecture was generalised to n CHEXs and $n-1$ PHEXs with A_{i+1} following B_i through PHEX i and $C_i = D_{i+1}$.

Decision: the inverse direction was left out

The inverse case, the "known A and B" CHEX solver and a check of required loads against a real cooler's capacity curves were discussed at length and later dropped as too complex, unnecessary or a tangent to what was needed. The forward model plus a *map* of the loads (Chapter 10) answers the practical question — what must the cryocooler provide — without them. An early version of the core did contain a `chex_AB` function; it was removed when the effectiveness definition changed.

The generalised state numbering was fixed then as well: CHEX i (warm to cold) has $A = 2i - 1$, $B = 2i$, and on the return side $C = 3n + 2 - i$, $D = 3n + 3 - i$; for $n = 4$ this gives the paper's 7, 8, 10, 11 for CHEX 4 and 1, 2, 13, 14 for CHEX 1. The valve inlet is $2n$, the outlet $2n + 1$, the evaporator outlet $2n + 2$. (A planning note once wrote the return-side index as $(2n + 2) - i$; that gives 6 for CHEX 4 and is wrong. The code had $3n + 2 - i$ from the start, kept in one helper, `state_labels(n)`.)

3.4 The core as a black box

The first code was a generic cycle solver, `jt_cycle_core.py`, with replaceable property backends. When the question came of reproducing the paper's plots, the answer was firm:

In the conversation

So let's use the main file as a blackbox, and let it output whatever results are required for this reproduction. Also for functions, we can find the error/correlation/closeness to the paper's functions. For speed of sound, let's use HEM since speed of sound data is not given even [in] CoolProp. All other reproductions etc. in one file.

Decision: separate the engine from the experiments

The cycle solver never knows about the paper. A separate script, `reproduce_zhang_dang_2021.py`, calls it, sweeps, plots, refits the paper's functional forms (Eqs. 35, 36, 49) and computes error metrics (RMSE, MAE, maximum and relative errors). Two kinds of result are never mixed: *reconstruction* (drawing the paper's own fitted curves from its coefficients) and *independent reproduction* (computing the same curves from properties). Only the second is evidence.

Decision: HEM instead of the paper's Γ method

The paper's choked flux needs the speed of sound of He-3, which was not available. A homogeneous-equilibrium model (HEM) finds the choked flux as the maximum of $G(P)$ along the isentrope, which needs only $h(P, s)$ and $\rho(P, s)$. It was kept explicitly as *our* model, distinct from the paper's, so that any difference in $\dot{m}$ could be attributed to the flow model and not to the cycle.

The early core had other features that later changed: the discharge coefficient fixed at 1, the mass flow *only* from an orifice diameter (a user-supplied $\dot{m}$ had been removed), vapour quality and liquid fraction reported at the throat, terminal-temperature checks on each CHEX, and a global

energy-balance residual. Ten unit tests on a synthetic ideal-gas fluid checked the logic — not He-3 physics, which was stated plainly at the time.

3.5 What was still missing

At that point the framework could redraw the paper's published fit curves (a "reference-only" mode) but could not compute a single He-3 number. The status list written then is a good snapshot of the project's honesty: "We have built the baseline cycle engine and the paper-reproduction framework… We have not yet demonstrated that the cycle model reproduces the paper's numerical results using helium-3 properties."

Chapter 4

The property problem

4.1 An offer from the professor

In the conversation

My professor asked me to make an Excel sheet requesting all the properties I require for Helium-3. He said I can ask for however many properties, even 10 000 won't matter. And then I can interpolate them and use that. So I have 2 major challenges: what properties to ask for; what points (T or P or something) to ask for. Need to firstly standardise the Excel sheet format. Then make a code which would use that standardised format and essentially be the CoolProp for He-3, the reason the whole thing is built to use CoolProp.

(This message had to be sent twice. The first time the assistant replied that properties would be "deferred". The lesson about stating the actual task plainly, and about the assistant listening to it, recurs later.)

4.2 What to ask for, and where

The request workbook had six sheets: instructions; a property catalogue; a P–T grid; saturation queries; constants and sources; and a "backend contract" describing exactly what the Python interpreter would need.

- **Essential:** T, P, ρ, h, s on a single-phase grid, and saturated-liquid and saturated-vapour values of the same. These are all the cycle and the HEM need: $h(P,T)$ and $T(P,h)$ for the exchangers, s and $h(P,s)$, $\rho(P,s)$ for the nozzle, P_{sat} and h_g for the evaporator, and the saturated endpoints for two-phase mixtures.
- **For the paper's Fig. 4:** c_p.
- **For later:** c_v, internal energy, compressibility, JT coefficient, speed of sound, viscosity, thermal conductivity (the last two only for a physical exchanger model that was never built).
- **Where:** a 100×100 grid over 0.3–350 K and 0.01 Pa–2 MPa, temperatures clustered at low T and near the critical point, pressures logarithmic with the operating pressures included; 602 saturation temperatures from 0.01 K up to the critical point, denser at low T and near criticality. The range below 1 K was chosen so the model could be pushed to $T_C = 0.5$ K.

Two cautions were recorded: the available He-3 equation of state covers 0.01–20 K only, so the warm end needs another source; and the critical constants in the literature (3.3157 K, 114.6 kPa) differ slightly from older values.

4.3 The interpreter: a CoolProp for He-3

The interpreter, `Helium3Fluid`, was written before the data existed, in the planning conversation (Chapter 5). Its rules were set then and never changed:

- never interpolate across the liquid–vapour jump: each isotherm is split at P_{sat}, and the saturation table adds an anchor node exactly on the boundary;
- two-phase states come from the saturation table, $h = h_l + x h_{fg}$, $1/\rho = (1-x)/\rho_l + x/\rho_v$;
- inverse lookups (T from h, h from s) root-find on the same interpolant, so forward and inverse

agree;
- no extrapolation: anything outside the table raises `OutOfRange`.

Because no He-3 data existed, it was tested by filling the same two CSV layouts with He-4 from CoolProp and comparing. On 1500 random off-grid states the median h error was 0.0016 J/kg (95th percentile 0.16 J/kg), ρ 6 $\times 10^{-7}$ relative, $T(P,h)$ 3 $\times 10^{-7}$ K; the worst point was next to the He-4 critical point (675 J/kg, 0.025 K at 5.16 K). A full cycle on the interpolated tables agreed with direct CoolProp to about 10^{-4} in q_C. That test proved the *code*, not the He-3 *physics*.

What actually came back from the data provider is described in Chapter 8: much coarser than requested, and with its own surprises.

Chapter 5

Planning in pseudo-code

A second, parallel conversation took the same paper and the first conversation's notes and asked a different question: before writing more Python, what exactly should the program do? It began with "Read the conversation. Understand the context fully" and moved quickly to "Let's focus more on pseudo code and code planning. We will translate into Python later."

5.1 Issues spotted in the paper and in the plan

The first reply listed five things to fix before building further:

1. the generalised A, B, C, D notation needed a clean mapping to the paper's 1–14;
2. the "known A and B" solver also needs the low-side pressure;
3. the paper's cold-end closure $T_8 \approx T_9 = T_{10}$ holds only near $\eta_4 = 1$; for $\eta_4 < 1$, h_8 must be computed from η_4;
4. the paper's Γ is stated inconsistently: Eq. 43 has $G = G_{ID}\Gamma$, but step (4) of the procedure divides G_{ID} by G. The physical form is $\Gamma = G/G_{ID}$; a solver that computes G directly from real-gas properties sidesteps the question;
5. mass flow and cycle seemed coupled (G depends on T_8, which depends on η_4), apparently needing a fixed-point iteration.

5.2 Two corrections that made the solver simple

The next message corrected item 5 and more:

Insight: the coupling is one-way

In the idealised model the enthalpy chain $h_{10} \to h_{11} \to \cdots \to h_{14}$ and $h_7 \to h_8 \to \ldots$ does not depend on $\dot{m}$. The flow is computed *after* the states:

$$T_C, P_D, T_{\text{pre}}, \eta \;\to\; \text{states} \;\to\; (T_8, P_U) \;\to\; G^* \;\to\; \dot{m} \;\to\; Q_C,\; Q_{\text{pre},i}.$$

No iteration is needed. Iteration would appear only with pressure drops, parasitic loads, or precooler temperatures that depend on their loads — none of which are in the model.

Insight: both CHEX closures are explicit

With $\eta = (h_D - h_C)/(h_{AD} - h_C)$ and $h_A - h_B = h_D - h_C$, no root-finding is needed for the exchanger. Every state of the cycle is one property call away from the previous one. A full solve takes milliseconds; most of the time goes into property inversions and the choked-flux search.

5.3 The layered plan

Layer	Content	Survived as
1 properties	$h, s, \rho, T(P,h)$, saturation	`jt/properties.py`, `CoolPropFluid`
2 components	CHEX solvers, valve, evaporator	`chex_limits`, `_chex_solve`, `chex_AC`
3 flow	critical throat, mass flux, $\dot{m}$	`critical_mass_flux_HEM`, `mass_flow`
4 cycle	states, loads, regime check	`solve_cycle`, `CycleResult`
5 applications	existing cooler; design	`jt/workflow.py` (partly)
6 entry	`run_jt_model(config)`	`jt_run.py`

Each layer imports only from those below it, and the property backend is passed in as an object so the cycle code never names a library. Physical checks are *flags, not silent fixes*: $\Delta h \geq 0$, $T_D \leq T_A$, $T_B \geq T_C$, $0 < \eta \leq 1$, and — written into the plan as a comment — "internal pinch check: sample intermediate enthalpies, confirm $T_{\text{hot}} \geq T_{\text{cold}}$ along the exchanger". That comment became a real function only later (Chapter 6).

The plan also proposed two applications: *evaluate an existing cooler* (prescribed T_{pre}, check each load against the cooler's capacity, treating capacities as inequalities so the problem is not over-determined) and *design to a target* T_C and Q_C (the orifice diameter is the free variable that closes the Q_C equality: $d_{\text{req}} = \sqrt{4\dot{m}_{\text{req}}/(\pi C_d G^*)}$ with $\dot{m}_{\text{req}} = Q_C/q_C$). The first was dropped in its capacity-curve form; what remained of both is the optimiser and the load map.

5.4 The validation plan

The validation targets written in this plan became, almost line for line, the scorecard that the finished code passes (Chapter 9):

1. CHEX algebra round-trips; global energy balance to machine precision;
2. ideal limit $q_C = h_{7D} - h_7$ (Eqs. 18–21);
3. optimum: $T_C = 1$ K, $T_{\text{pre3}} = 8$ K, $P_D = 1.16$ kPa $\Rightarrow P_{U,\text{opt}} \approx 0.36$ MPa, $q_C \approx 8.0$ kJ/kg, $\Delta h_T \approx q_L$;
4. $P_{U,\max} = 1.13$ MPa;
5. mass flux: $G \approx 7450\ \text{kg m}^{-2}\text{s}^{-1}$ from $\dot{m} = 2.34$ mg/s through 20 µm; Eq. 49 gives about 7460 if Table 2's γ is read as -0.58×10^3; $\Gamma \approx 1.5$;
6. $Q_C \approx 18.7$ mW at $\eta_4 = 1$ and 14.7 mW at $\eta_4 = 0.97$;
7. about 7 % loss of Q_C per 1 % of η_4.

"Tests 6–8 are the hard ones. They depend on the He-3 equation of state and on how the choke search treats the two-phase region."

5.5 Five open decisions, and how each was settled

Open question (planning)	How it ended
Choke model in the liquid/two-phase region: equilibrium or frozen sound speed, or the paper's single-phase EOS procedure?	Equilibrium (HEM), by maximising G; no sound speed needed. At 1 K and 0.36 MPa the inlet is liquid-like and the HEM maximum sits at the back pressure (flag `UNCHOKED_RESTRICTION_FLOW`); the flux still matches the paper to $< 1\,\%$.
Over-cooled regime for $\eta_4 < 1$: clip q_C at q_L, or reject the point?	Neither: redefine the effectiveness against both exchanger limits, so the regime cannot be entered (Chapter 6).
P_D: always $P_{\text{sat}}(T_C)$, or an input?	Always $P_{\text{sat}}(T_C)$. This later limited the closed-loop Ma case (Chapter 11).
Design objective?	Q_C for the last stage (with a "practical" variant), Carnot-weighted precooling load for the upstream temperatures (Chapter 10).
Cooler capacity data format?	Not needed: replaced by the load map.

5.6 A moment of frustration worth remembering

At the end of that conversation, after a property interpreter, a CoolProp test harness, a request workbook and new versions of every file had appeared:

In the conversation

In all honesty I can't really read through the code to make sure if it is written correctly. I need this project in my resume, for which what code I used isn't a priority if it matches the approach I discussed. So please make sure it does, and kinda explain me as well. You honestly tend to overdo things sometimes which is very bad for me. Also, how does all these files tie together? How do I run them?

Insight: the requirement behind every later choice

This is the real specification of the project: *the code must implement the approach I worked out, I must be able to run it, and I must be able to explain it*. Everything that followed — the single front-door script, the scorecard, the feasibility messages written in words, this journal — is an answer to that paragraph.

Chapter 6

The review, and the bug in the effectiveness

The long working session that produced the present code began like this:

In the conversation

These are all the project files so far. Also the original research papers. The CSVs for helium-3 aren't yet available, everything else is. Read all this, then I'll ask you to make things I want. … Any issues you see?

The files were the core solver, the He-3 interpreter, a last-stage optimiser ("phase 1": grid search plus refinement of Q_C over P_U and $T_{\text{pre,last}}$, using either the paper's Eq. 49 or HEM, writing results under names like `Helium_3_2.1K_20um_4_stage`), a PHEX load map ("phase 2": fix that optimum and $\dot{m}$, sweep the warmer precooler temperatures, report every load), the reproduction script and two test files.

6.1 The serious issue

Because there was still no He-3 data, the suspected problem was tested on He-4 from CoolProp. The last-stage optimiser evaluated Eq. 5 of the paper literally, $q_C = h_{10} - h_7 + \eta(h_{7D} - h_{10})$, and set $h_8 = h_{10} - q_C$. At $\eta = 1$ that is $q_C = \Delta h_T$ *always* — also above the optimum pressure, where the paper itself says $q_C = q_L$.

What went wrong: the optimiser accepted impossible operating points

He-4, $T_C = 2.6$ K, $T_{\text{pre}} = 6$ K, $\eta = 1$, HEM flow:

P_U [MPa]	T_8 [K]	What happened
0.3	5.0	accepted, 22.9 mW (valid)
0.4	2.31, *below* $T_C = 2.6$ K	accepted silently, 70.9 mW — about 3× its neighbour
≥ 0.5	—	property backend error (h_8 below the table)

The gas leaving the last CHEX was colder than the evaporator it came back from. The result depended on how far below T_C the property table happened to extend: with a table going well below T_C, the optimiser could run to the upper pressure bound and report a large, meaningless Q_C. The full-cycle solver had the matching symptom: above the optimum at $\eta = 1$ it crashed in a property inversion instead of raising the `COLD_END_TEMPERATURE_CROSSOVER` flag it had been written to produce.

The cause is the fictitious state A' of handwritten page 2. The effectiveness was defined against the cold-side maximum $h_{A'} - h_C$ only. When the cold end is the one that runs out of temperature difference, that "maximum" is not a maximum at all.

6.2 The fix: an exchanger has two limits

The fix was to ask, from scratch, how much heat a counterflow exchanger *can* transfer between two given inlet streams. Two things can stop it:

- the cold stream cannot leave warmer than the hot stream enters: $h_D \leq h(P_D, T_A)$, so $Q \leq$

$W_{\text{warm}} = h(P_D, T_A) - h_C$ (this is $h_{A'} - h_C$);
- the hot stream cannot leave colder than the cold stream enters: $h_B \geq h(P_U, T_C)$, so $Q \leq W_{\text{cold}} = h_A - h(P_U, T_C)$.

$Q_{\max} = \min(W_{\text{warm}}, W_{\text{cold}})$ and $\eta = Q/Q_{\max}$. Then $h_B = h_A - \eta Q_{\max}$, $h_D = h_C + \eta Q_{\max}$, and any $\eta \in [0, 1]$ is physically admissible by construction.

Insight: the minimum rule is not coded — it falls out

For the last CHEX, $q_C = h_{10} - h_8 = (h_{10} - h_7) + Q$. Substituting the two limits:

$$(h_{10} - h_7) + W_{\text{warm}} = h_{7D} - h_7 = \Delta h_T, \qquad (h_{10} - h_7) + W_{\text{cold}} = h_{10} - h(P_U, T_C) = q_L.$$

So with $\eta = 1$, $q_C = \min(\Delta h_T, q_L)$ *automatically*. The quality argument of handwritten page 5 and the exchanger argument are the same statement. And for $\eta < 1$:

$$q_C = \eta \, \min(\Delta h_T, q_L) - (1 - \eta)(h_7 - h_{10}),$$

the formula behind the large sensitivity to η (Chapter 9).

Insight: the optimum is where both ends pinch

This sentence from the session was saved separately because it captures the whole last stage:

In the conversation

The optimum is where the two limits meet. Only there is $T_8 = T_C$ and the cold stream leaves at T_{pre}.

Below the optimum pressure the last CHEX is warm-end limited (return gas leaves at T_{pre}, gas before the valve is warmer than T_C); above it, cold-end limited (gas before the valve at T_C, return gas leaves colder than T_{pre}). At $\Delta h_T = q_L$ both happen at once. The handwritten pinch argument of page 3 — cold-end pinch when $c_{p,U} < c_{p,D}$, hot-end pinch otherwise — is this, seen through heat capacities.

Decision: keep the paper's numbers, change its definition

The paper's effectiveness equals Q/W_{warm}. Whenever the warm end binds — which includes the paper's last CHEX at and below its optimum, the regime of its whole analysis — this is identical to $Q/Q_{\max}$. So redefining η changes no number in the paper, removes the unphysical branch, and makes η comparable with the effectiveness other authors quote from measured temperatures. The code says so in `chex_AC`'s docstring, and `chex_limits` reports which end binds for every CHEX of every solve.

6.3 The other findings

- **T_8 versus the paper's $T_0 \approx T_C$.** The paper evaluates the flux with the valve inlet at T_C. Below the optimum pressure, or with $\eta < 1$, the real T_8 is warmer. We kept the real T_8; it is why $\dot{m}$ falls as η falls, as in the paper's Fig. 15. It also meant `paper_fit` (valid only for T_8 of 1–3 K and 0.2–2 MPa) would reject warm points, which showed up as "failed evaluations" rather than answers.
- **No internal pinch check.** Only the end temperatures of each CHEX were checked, although the paper itself discusses $c_{p,U} < c_{p,D}$ at the cold end.
- **A test that could not fail.** The interpreter test against CoolProp printed errors but asserted nothing. Its He-4 table also ran up to 5.19 K while He-3 critical constants (3.3246 K) were hard-coded, so above 3.3 K it exercised branches a real He-3 table never would.

- **Smaller points:** c_p as a PCHIP derivative is noisy; the strict monotonicity check could fire spuriously near saturation; the HEM grid search (101×101 points, each with a 180-point flux sweep) was slow; `requirements.txt` lacked matplotlib and pinned odd versions.

6.4 The internal pinch check

A counterflow exchanger can satisfy both end conditions and still have the hot stream colder than the cold stream somewhere inside, when c_p varies strongly along it. The check that was built uses a small trick: because both streams carry the same flow, *the heat already exchanged* labels a position along the exchanger. Counting q from the cold end,

$$h_{\text{hot}}(q) = h_B + q \ \ (\text{at } P_U), \qquad h_{\text{cold}}(q) = h_C + q \ \ (\text{at } P_D), \qquad \Delta T(q) = T(P_U, h_{\text{hot}}) - T(P_D, h_{\text{cold}}).$$

Twenty equal steps in q from 0 to Q, only $h \to T$ inversions, no c_p, no geometry. A negative gap raises `INTERNAL_TEMPERATURE_CROSSOVER`; an interior gap smaller than both end gaps raises `INTERNAL_PINCH`. It is a flag and never changes a result.

Insight: what it found

In the paper's own design, CHEX 1 shows a crossover of about -0.01 K at 10–30 % of its duty. It is a hint that an ideal ($\eta = 1$) warm exchanger is marginal there, not a fault in the solver; the paper's numbers are unaffected because CHEX 1 cannot influence q_C (Chapter 10).

6.5 What changed in the code

Comparing the core as uploaded with the core as it is now:

Before	After
`chex_AC`: $h_D = h_C + \eta(h(P_D, T_A) - h_C)$ (cold side only)	`chex_limits` computes W_{warm}, W_{cold} and the binding end; `_chex_solve` uses $Q = \eta \max(Q_{\max}, 0)$
`chex_AB` (inverse)	removed
`chex_terminal_flags` only	plus `chex_profile_check` (20-point interior scan)
optimiser: Eq. 5 literally, $h_8 = h_{10} - q_C$	thin wrapper over `solve_cycle` with one CHEX, so it obeys the same limits
$\dot{m}$ only from an orifice	orifice *or* measured `mdot_mg_s` (Chapter 11)
HEM: no tolerance on round-off at the stagnation point	tolerance of 10^{-3} J/kg on $h_0 - h(P_0, s_0)$, after real tables showed the round-trip noise (Chapter 8)

Chapter 7

The ideas behind the solver, explained slowly

Chapter 6 told what was found and fixed. This chapter goes back over the four ideas that took the longest to understand, in the order in which they finally made sense: what "effectiveness" really means, why q_C is a minimum and why that minimum is actually reached, what an internal pinch is, and how the choked flow through the valve is computed when the paper's method cannot be used. The numbers are all for the paper's last stage: He-3, $T_C = 1$ K, $T_{\text{pre3}} = 8$ K, from the tables.

7.1 Effectiveness: one word, three definitions

The textbook idea

A counterflow exchanger takes two streams and moves heat from the hot one to the cold one. Its *effectiveness* compares what it does with what the best possible exchanger — infinitely long, same inlets — would do:

$$\eta = \frac{Q}{Q_{\max}}.$$

The whole question is what $Q_{\max}$ is. In a textbook with constant heat capacities, the answer is $Q_{\max} = C_{\min}(T_{h,\text{in}} - T_{c,\text{in}})$: an infinitely long exchanger heats or cools the stream with the *smaller* heat-capacity rate $C = \dot{m}c_p$ all the way to the other stream's inlet temperature, and then it has to stop, because that stream has no temperature difference left. The other stream changes temperature less, because it has more heat capacity.

The same idea with real fluids

With He-3 between 1 and 8 K, c_p is nowhere near constant (Fig. 2.4): at 0.36 MPa it falls from about 12 kJ/kg/K near 5 K to about 1.3 kJ/kg/K at 1 K, while at 1.16 kPa it stays near 7. There is no single $C_{\min}$. But the idea survives if it is written in enthalpies. In an infinitely long exchanger, one of two things stops the heat transfer:

- **The cold stream reaches the hot inlet temperature** at the warm end ($T_D \to T_A$). The heat it has picked up is then
$$W_{\text{warm}} = h(P_D, T_A) - h_C.$$
This is computed on the *low-pressure* isobar. It is the handwritten $h_{A'} - h_C$.
- **The hot stream reaches the cold inlet temperature** at the cold end ($T_B \to T_C$). The heat it has given up is then
$$W_{\text{cold}} = h_A - h(P_U, T_C).$$
This is computed on the *high-pressure* isobar.

Both streams carry the same flow and exchange the same heat, so the exchanger stops at whichever of the two comes first:

$$Q_{\max} = \min(W_{\text{warm}}, W_{\text{cold}}).$$

The "stream with the smaller heat capacity" of the textbook is exactly the stream whose limit is smaller; the enthalpy form just does not need an average c_p to find it.

The three definitions

Definition	When it is right
Cold-side: $\eta = Q/W_{\text{warm}} = (h_D - h_C)/(h(P_D, T_A) - h_C)$. The paper's Eq. 4 and handwritten page 2.	Whenever the warm end limits. Then W_{warm} *is* $Q_{\max}$.
Hot-side: $\eta = Q/W_{\text{cold}} = (h_A - h_B)/(h_A - h(P_U, T_C))$.	Whenever the cold end limits. Used when a paper quotes the gas temperature before the valve.
General: $\eta = Q/\min(W_{\text{warm}}, W_{\text{cold}})$. Used by the solver.	Always. Reduces to one of the other two depending on which end binds.

What happens if the wrong one is used? Take the paper's last CHEX at three pressures:

P_U [MPa]	W_{warm} [kJ/kg]	W_{cold} [kJ/kg]	binding end	q_C at $\eta = 1$ [kJ/kg]
0.25	48.64	52.30	warm	$\Delta h_T = 5.53$
0.358	48.64	48.64	both	7.98
0.60	48.64	41.04	cold	$q_L = 5.39$

W_{warm} does not depend on P_U at all (it is on the low-pressure isobar). W_{cold} falls as P_U rises. At 0.6 MPa the cold-side definition with $\eta = 1$ asks the exchanger to transfer 48.64 kJ/kg when only 41.04 kJ/kg can be taken from the hot stream before it reaches 1 K. The extra 7.6 kJ/kg can only come from cooling the gas *below* 1 K, colder than the stream cooling it. That is the bug of Chapter 6, in numbers.

7.2 Why $q_C = \min(\Delta h_T, q_L)$, and why the minimum is reached

Two ceilings

For the last CHEX, the refrigeration is what the evaporator absorbs: $q_C = h_{10} - h_9 = h_{10} - h_8$. The exchanger takes Q from the supply gas, so $h_8 = h_7 - Q$ and

$$q_C = (h_{10} - h_7) + Q.$$

$h_{10} - h_7$ is fixed by the inlets (it is negative: the 8 K gas has more enthalpy than the 1 K vapour, by 40.7 kJ/kg at the optimum). So *the refrigeration is the heat the last exchanger manages to move, minus a fixed amount*. Every extra joule the exchanger transfers is a joule of refrigeration. Now put in the two limits:

$$q_C \le (h_{10} - h_7) + W_{\text{warm}} = h(P_D, T_7) - h(P_U, T_7) = \Delta h_T,$$
$$q_C \le (h_{10} - h_7) + W_{\text{cold}} = h_g(T_C) - h(P_U, T_C) = q_L.$$

q_C sits under two ceilings. Whatever the exchanger does, it is below both, so below the lower one: $q_C \le \min(\Delta h_T, q_L)$.

Why the lower ceiling is actually touched

A bound is only a bound. What makes it an equality is the exchanger: with $\eta = 1$, $Q = Q_{\max} = \min(W_{\text{warm}}, W_{\text{cold}})$, so q_C is pushed up until it hits whichever ceiling is lower. Physically, an ideal

counterflow exchanger keeps transferring heat until one end runs out of temperature difference, and *which end runs out first is the same thing as which ceiling is lower*:

- **Below the optimum pressure** ($\Delta h_T < q_L$): the warm end pinches. The return vapour leaves at T_{pre}, carrying away exactly as much enthalpy as it can; the gas before the valve is still warmer than T_C; after throttling it is a vapour–liquid mixture that does not have enough liquid to absorb more. The cooling is limited by the JT effect available at T_{pre}: $q_C = \Delta h_T$.
- **Above the optimum pressure** ($\Delta h_T > q_L$): the cold end pinches. The gas before the valve is cooled all the way to T_C; after throttling there is as much liquid as there can be. More JT effect is available but cannot be used: the return vapour leaves the exchanger *colder* than T_{pre}, i.e. the unused refrigeration is carried back up to the precooler and wasted. The cooling is limited by the latent heat: $q_C = q_L$.
- **At the optimum** both ends pinch at once: $T_8 = T_C$ and the return vapour leaves at T_{pre}. All of the JT effect is turned into liquid and all of the liquid is evaporated. Nothing is wasted at either end.

Insight: the realisation

$q_C = \min(\Delta h_T, q_L)$ is not a rule about the evaporator; it is the statement that the last exchanger can only move as much heat as its tighter end allows, and that every bit of heat it moves becomes refrigeration. The handwritten quality argument ("you cannot evaporate more than all the liquid") is the cold-end half of this; the warm-end half is that you cannot use more JT effect than exists at T_{pre}. The optimum pressure is the one at which the two halves balance.

Why the two trends cross

$\Delta h_T = \int_{P_D}^{P_U} v(\beta T - 1)\,\mathrm{d}P$ at T_{pre} grows with P_U as long as $\beta T > 1$ (below the inversion temperature). $q_L = h_g - h(P_U, T_C)$ falls with P_U because the compressed liquid's enthalpy rises with pressure, $(\partial h/\partial P)_T = v(1 - \beta T) > 0$ for a liquid. One rises and one falls; the minimum of the two is largest where they cross. q_L reaches zero at $P_{U,\max} = 1.13$ MPa: compressed to that pressure, the liquid at 1 K already has the enthalpy of saturated vapour, and throttling it produces no liquid at all.

With $\eta < 1$

$Q = \eta\, Q_{\max}$ gives $q_C = \eta \min(\Delta h_T, q_L) - (1 - \eta)(h_7 - h_{10})$. The minimum is no longer reached; the exchanger stops short of the tighter end by a fraction $1 - \eta$ of $Q_{\max} \approx 48.6$ kJ/kg, which is six times q_C. Hence the steep loss with η.

7.3 Internal pinch

The two limits above are conditions at the *ends* of the exchanger. They guarantee $T_B \geq T_C$ and $T_D \leq T_A$. They do not guarantee that the hot stream is warmer than the cold stream *everywhere inside*. If c_p of one stream changes strongly along the exchanger, its temperature profile can bend so that the two profiles touch, or cross, somewhere in the middle, while both ends still have a positive difference. That is an *internal pinch*; a crossing is physically impossible (heat would have to flow from cold to hot), and a touch means the true maximum heat transfer is smaller than $\min(W_{\text{warm}}, W_{\text{cold}})$.

How to see inside without geometry

The trick used in `chex_profile_check`: since both streams have the same mass flow, at any cross-section the heat the cold stream has absorbed so far, counted from the cold end, equals the heat the hot stream has given up from that point to the cold end. So the heat q labels the position, and

$$h_{\text{hot}}(q) = h_B + q \;\; (P_U), \qquad h_{\text{cold}}(q) = h_C + q \;\; (P_D), \qquad 0 \le q \le Q.$$

Converting each enthalpy to a temperature on its isobar gives both profiles as functions of q, from T_B and T_C at $q = 0$ to T_A and T_D at $q = Q$. The code evaluates them at 21 points and looks at $\Delta T(q) = T_{\text{hot}} - T_{\text{cold}}$:

- $\Delta T < -10^{-3}$ K somewhere $\Rightarrow$ `INTERNAL_TEMPERATURE_CROSSOVER`: this exchanger, with this η, is not physically possible;
- the smallest interior ΔT below both end values $\Rightarrow$ `INTERNAL_PINCH`: the tightest point is inside, not at an end.

No c_p, no length, no area — only the $h \to T$ inversion the solver already has. It is a check, not a correction: the result is never changed, the flag tells you to distrust it.

Insight: what the flag on CHEX 1 means

The paper's warmest exchanger (300 K down to about 26 K at $\eta = 1$) shows a crossover of about -0.01 K. The end limits say an ideal CHEX 1 is fine; the profile says it would need a hair less than ideal. Since CHEX 1 cannot affect q_C, $\dot{m}$ or Q_C (Chapter 10), the paper's results stand; but a real design would use η_1 slightly below 1, which the solver allows.

7.4 Choked flow: why not the paper's method

What choking is

Gas or liquid accelerating through a restriction converts enthalpy into kinetic energy. The mass flux $G = \rho u$ through the narrowest section first rises as the downstream pressure is lowered, then stops rising: once the flow at the throat reaches the local speed of sound, pressure changes downstream can no longer travel upstream to "tell" the throat, and G is fixed. That limiting flux is the choked flux.

The paper's route

Zhang & Dang (Sect. 4, Eqs. 41–48) follow Maytal's real-gas procedure: start from the ideal-gas choked flux G_{ID} (a closed formula in P_0, T_0, γ, M), find the real throat state from $s^* = s_0$ and $h^* + a^{*2}/2 = h_0$ with a real-gas equation of state, and express the result as a deviation factor $\Gamma = G/G_{ID}$, then fit $G = \alpha P_U^{\beta} + \gamma$ (Eq. 49, Table 2). Handwritten pages 7–8 reproduce this.

Why it was hard to use here

1. **It needs the speed of sound $a(P, T)$ of He-3.** The solver's properties come from tables; the tables that came back contain h, s and ρ only, and CoolProp has no He-3 at all, so there was no independent source.
2. **In the saturated region the speed of sound is not even well defined.** The expansion through the valve starts from a compressed liquid near T_C and may enter the two-phase dome. Inside the

dome, the sound speed depends on what one assumes about the phases: if liquid and vapour stay in equilibrium ("equilibrium sound speed") it is very low and drops discontinuously at the saturation line; if they do not have time to exchange heat and mass ("frozen") it is much higher. The paper's sonic condition $V^* = a$ therefore requires a modelling choice the paper does not spell out, and a single-phase equation of state does not make it.

3. Γ **is stated inconsistently** (Chapter 5): $G = G_{ID}\Gamma$ in Eq. 43, G_{ID}/G in the procedure.
4. **The fit is narrow**: Eq. 49 exists only for $T_C = 1, 2, 2.6, 3$ K, 0.2–2 MPa, and evaluates the flux at $T_0 = T_C$, which is not the real T_8 when $\eta < 1$.

7.5 HEM: theory

The *homogeneous equilibrium model* makes two assumptions about the flow through the restriction: the phases move together at one velocity (*homogeneous*), and they are always in thermal and phase equilibrium at the local pressure (*equilibrium*). Then the two-phase mixture behaves like a single fluid with mixture properties, and the flow is adiabatic and, ideally, isentropic. From stagnation state (P_0, h_0, s_0), at any downstream pressure P on the isentrope:

$$h = h(P, s_0), \qquad u = \sqrt{2(h_0 - h)}, \qquad \rho = \rho(P, s_0), \qquad G(P) = \rho\, u.$$

If s_0 lies between the saturated-liquid and saturated-vapour entropies at P, the state is two-phase:

$$x = \frac{s_0 - s_l}{s_v - s_l}, \qquad h = h_l + x(h_v - h_l), \qquad \frac{1}{\rho} = \frac{1-x}{\rho_l} + \frac{x}{\rho_v}.$$

Why the maximum of G is the choked flux

At $P = P_0$, $u = 0$ so $G = 0$. As P falls, u rises but ρ falls. Along the isentrope $\mathrm{d}h = \mathrm{d}P/\rho$, so $u\,\mathrm{d}u = -\mathrm{d}P/\rho$, and

$$\frac{\mathrm{d}G}{\mathrm{d}P} = u\frac{\mathrm{d}\rho}{\mathrm{d}P} + \rho\frac{\mathrm{d}u}{\mathrm{d}P} = u\frac{\mathrm{d}\rho}{\mathrm{d}P} - \frac{1}{u} \quad \Longrightarrow \quad \frac{\mathrm{d}G}{\mathrm{d}P} = 0 \iff u^2 = \left(\frac{\partial P}{\partial \rho}\right)_s = a^2.$$

The pressure at which G is largest is exactly the pressure at which the flow speed equals the (equilibrium) speed of sound. So *maximising* $G(P)$ along the isentrope finds the sonic throat *without ever computing a*. The equilibrium assumption of HEM is what decides, implicitly, which sound speed is meant in the dome.

- If the maximum lies between P_D and P_0, the flow is choked: $G = G_{\max}$ and lowering P_D further changes nothing.
- If G is still rising at the back pressure P_D, the flow never reaches sonic speed: it is *not choked* and $G = G(P_D)$.

What happens at 1 K

At the paper's point the stagnation state is liquid He-3 at 0.358 MPa and 1 K. Following its isentrope down:

P [kPa]	ρ [kg/m^3]	u [m/s]	G [kg m^{-2}s^{-1}]
300	89.6	36.0	3223
100	84.9	76.7	6507
20	82.5	88.3	7277
2	81.8	90.7	7423
1.160 ($= P_D$)	81.8	90.8	7430

The density hardly changes: it is a liquid, and its isentrope stays liquid all the way to $P_{\text{sat}}(1\,\text{K}) = P_D$. G keeps rising to the back pressure, so the flow is not choked (the code says `UNCHOKED_RESTRICTION_FLOW`), and the flux is essentially Bernoulli flow of a liquid, $G \approx \sqrt{2\rho\Delta P}$. The result, $G = 7430$ and $\dot{m} = 2.334$ mg/s through 20 µm, is within 0.3 % of the paper's Table 2 and its 2.34 mg/s. With $\eta < 1$, T_8 rises, the liquid is less dense and nearer to flashing, and G falls — the drop in $\dot{m}$ of the paper's Fig. 15.

7.6 HEM: how the code does it

`critical_mass_flux_HEM(P0, T0, P_back, fluid, n_grid=500)` in `jt/cycle_core.py`:

1. **Stagnation state.** $h_0 = h(P_0, T_0)$, $s_0 = s(P_0, T_0)$, with $(P_0, T_0) = (P_U, T_8)$.
2. **One evaluation** `evaluate(P)`: ask the property backend for $h(P, s_0)$ and $\rho(P, s_0)$. In `Helium3Fluid` this first checks the saturation table at P: if $s_l < s_0 < s_v$ the state is two-phase and the mixture formulas above are used; otherwise it builds the isobar profile at P and root-finds $s(T) = s_0$ (`brentq`) for T, then reads h and ρ. Then $u = \sqrt{2(h_0 - h)}$, $G = \rho u$, and the vapour quality for reporting. Interpolation round-off can make $h(P_0, s_0)$ differ from h_0 by a fraction of a J/kg, so negative $h_0 - h$ down to 10^{-3} J/kg is treated as zero.
3. **Coarse search.** 501 pressures spaced logarithmically from P_{back} to P_0 — logarithmic because the range spans a factor of 300 (1.16 kPa to 0.36 MPa) and the interesting part can be anywhere. Take the largest G.
4. **Decide.** Maximum at the first point (P_{back}): not choked, $P^* = P_{\text{back}}$. Maximum at P_0: impossible ($G(P_0) = 0$), so raise an error — it means the property data are wrong. Otherwise: choked, refine.
5. **Refine.** Golden-section search between the two grid neighbours of the best point, 80 iterations, no derivatives (derivatives of interpolated tables are noisy). Keep the best of the refined point and its neighbours.
6. **Report.** G, P^*, ρ^*, u^*, choked or not, phase at the throat (single-phase, saturated, two-phase) with quality and liquid fraction. If the backend has a sound speed (CoolProp fluids do; He-3 tables do not), $|u^* - a^*|/a^*$ is reported as a diagnostic that the maximum really is sonic.

`mass_flow` multiplies by the orifice area, $\dot{m} = \frac{\pi}{4} d^2 G$ ($C_d = 1$). Each evaluation costs two property inversions, so one flux costs about a thousand; that is why scans of Q_C over $(P_U, T_{\text{pre,last}})$ are the slow part of the optimiser, and why the optimiser exposes a coarser `--hem-grid`. The paper's own Eq. 49 remains available as `--flow-model paper_fit` for comparison, inside its range.

Decision: HEM, and say so

HEM needs only h, s, ρ and saturation data — exactly what the tables contain — works for any fluid in CoolProp too, handles the dome by an explicit assumption (equilibrium) instead of an implicit one, and reproduces the paper's flux to better than 1 % from 1 to 3 K. It is a different method from the paper's, and every report says which one was used.

Chapter 8

When the He-3 data arrived

8.1 What came back

Two CSV files arrived: a single-phase table and a saturation table. They were not the 100×100 grid that had been requested.

	Requested	Received
Single-phase grid	100 T × 100 P	67 T × 59 P (3953 points)
Temperature range	0.3–350 K	0.5–300 K (38 isotherms below 3.33 K, 19 between 3.33 and 20 K)
Pressure range	0.01 Pa–2 MPa	1 Pa–3 MPa, logarithmic
Columns	T, P, ρ, h, s (+ c_p and others)	T, P, h, s, ρ
Saturation	602 points, 0.01 K to T_{crit}	30 points, 0.5–3.2 K (0.1 K steps, 0.05 K above 3 K)

The headers were also not the ones the loader expected (`h liq [J/kg]`, `s vap [J/kg/K]`, `P [Pa]` instead of `h_liquid`, `P_sat`, ...). The files now live in `data/he3/` as `SinglePhase_He_3.csv` and `Saturation_He_3.csv`.

8.2 Four problems the real tables exposed

What went wrong: things a synthetic test could never have shown

1. **Headers.** Fixed by normalising column names and accepting known alternatives, rather than asking for the files to be edited.
2. **The saturation table stops at 3.2 K, below the critical point.** The interpreter decides liquid side versus vapour side by comparing with $P_{\text{sat}}(T)$; above 3.2 K it had no curve. The critical point (3.3246 K, 116.4 kPa, from the 1962 He-3 temperature scale; $P_{\text{crit}} \approx 873$ mmHg) is appended to the T–P_{sat} curve *for classification only*: no saturated h, s or ρ is invented above 3.2 K. (The literature value quoted when the data were requested, 3.3157 K and 114.6 kPa, is slightly different; it is used for nothing.)
3. **A near-critical gap.** Between the last tabulated P_{sat} (101.7 kPa) and P_{crit} there is no saturation data at all. Requests there raise `OutOfRange`. A Q_C-versus-P_U sweep once failed exactly there, at 0.1023 MPa; such sweeps now start at 0.12 MPa and record failures as missing points.
4. **Round-off at edges.** An $\eta = 1$ warm exchanger returns gas at exactly 300 K, the edge of the table; the inversion missed it by round-off. Targets within 10^{-9} (relative) of a table edge are now accepted. Likewise $h(P_0, s(P_0, T_0))$ does not return h_0 exactly; the HEM now tolerates 10^{-3} J/kg (a velocity of a few cm/s).

One more property of the real grid matters in practice: the liquid side of an isotherm needs at least three nodes for PCHIP, so liquid states below about 45 Pa ($T \lesssim 0.55$ K) are unavailable. T_C is therefore usable from about 0.55 K to 3.2 K.

8.3 How the interpolation works now

For a request (P, T):

1. Decide the class: liquid side ($P > P_{\mathrm{sat}}(T)$ or $P \geq P_{\mathrm{crit}}$) or vapour side.
2. On every tabulated isotherm of that class, interpolate h, s and $\ln \rho$ in $\ln P$ to the requested pressure (monotone PCHIP), adding the saturation anchor where the isotherm crosses P_{sat}.
3. The resulting values form an *isobar profile* in T; interpolate it (PCHIP again) at the requested T. Inverse requests (T from h, from s) root-find (`brentq`) on this same profile.
4. Two-phase: from the saturation table, $x = (s - s_l)/(s_v - s_l)$ etc.

A hold-out test (dropping every other grid line and predicting it) gave a median $|\Delta h|$ below 2 J/kg in the cycle's range, a 95th percentile of tens of J/kg and a worst case of a few hundred J/kg near the critical point and between 5 and 13 K — where c_p at high pressure has its peak (Fig. 2.4). The real error with the full grid is smaller than this, since the hold-out doubles the spacing.

8.4 The data check

`python jt_run.py check` runs before anything else and prints, for the supplied tables:

files found	SHA-256 `b17901e3e299 / 8a674a6a03da`
loader	columns, units, monotonicity, saturation split accepted
ranges	0.5–300 K, 1 Pa–3 MPa; saturation 0.5–3.2 K
$P_{\mathrm{sat}}(1\ \mathrm{K})$	1160.3 Pa (paper: about 1.16 kPa)
round trip	$h(0.36\ \mathrm{MPa}, 8\ \mathrm{K}) \to T$ returns 8.00000 K
not checked	the reference state of h and s (only differences enter any result)

Decision: fail loudly, never extrapolate

Every one of the problems above could have been "solved" by extrapolating. Each time the choice was to raise an error with the offending T and P instead. A number that comes back from the solver is therefore always an interpolation inside the supplied data.

Chapter 9

Reproducing Zhang & Dang

9.1 How the reproduction is organised

`python jt_run.py validate` runs four steps and stops at the first that fails: the data check, the unit tests, the reproduction, and a scorecard. The reproduction (`reproduce_zhang_dang_2021.py`) has five sections, one per group of the paper's results:

Section	What it computes
`fits`	$P_{U,\mathrm{opt}}$ and $q_{C,\mathrm{opt}}$ for a range of T_{pre3}; compare with Eqs. 35–36
`fig5`	Δh_T, q_L, q_C against P_U at $T_C = 1$ K, $T_{\mathrm{pre3}} = 8$ K
`fig7`	q_C against T_{pre3} for several P_U; $T_{\mathrm{pre3,opt}}(P_U)$ against Eq. 36
`flux`	HEM mass flux against P_U at $T_C = 1, 2, 2.6, 3$ K; compare with Table 2/Eq. 49 and refit Eq. 49's form to our data
`fig15`	q_C, $\dot{m}$, Q_C against η_4

Decision: three kinds of check, kept apart

The scorecard labels each line. *Anchors* compare with a number printed in the paper. *Consistency* checks test our implementation against its own identities ($q_C = \min(\Delta h_T, q_L)$ to 10^{-6}; $\Delta h_T = q_L$ at the optimum; no optimum on a search bound; Eq. 49's form fits our flux data) — these must pass whatever the paper says. *Model* checks compare things that depend on our modelling choices (HEM against the paper's Γ flux; $\dot{m}$; Q_C); a miss there is only a warning. This separation is what lets a failure be diagnosed rather than just noticed.

9.2 The result

27 of 27 checks pass. The headline numbers:

Quantity	Solver	Paper	Kind
$P_{U,\mathrm{opt}}$ at $T_{\mathrm{pre3}} = 8$ K	0.3584 MPa	0.36 MPa	anchor
$q_{C,\mathrm{opt}}$	7.979 kJ/kg	8.0 kJ/kg	anchor
$P_{U,\max}$ ($q_L = 0$)	1.131 MPa	1.13 MPa	anchor
$q_C - \min(\Delta h_T, q_L)$	4×10^{-14}	—	consistency
$P_{U,\mathrm{opt}}$ vs Eq. 36, mean / worst	1.39 % / 3.72 %	—	anchor
$q_{C,\mathrm{opt}}$ vs Eqs. 35+36, mean	3.12 %	—	anchor
HEM flux vs Table 2, mean, $T_C = 1$–3 K	0.25–0.69 %	—	model
$\dot{m}$, Q_C at $\eta_4 = 1$	2.339 mg/s, 18.62 mW	2.34, 18.7	model
Q_C at $\eta_4 = 0.97$; drop	14.74 mW; 20.8 %	14.7; about 21 %	model / anchor

(These are the dense-grid values, `validate --dense`; Chapter 13. The quick grid on my laptop gave 1.65 % and 3.44 % for the two fit means and 0.24–0.82 % for the flux, and the full grid 1.5 % and 3.3 %. Every pass/fail is the same on all three grids.)

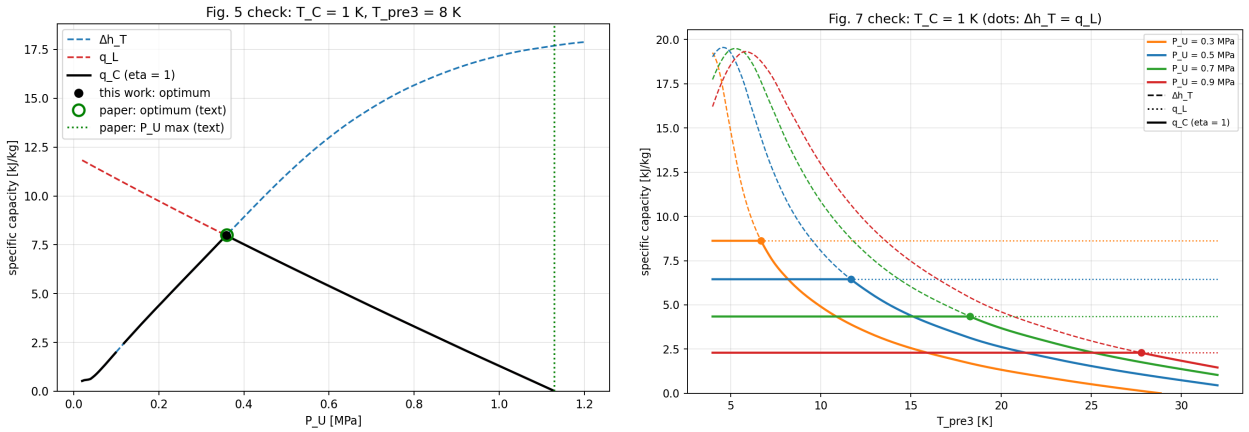


Figure 9.1: Our versions of the paper's Fig. 5 (left) and Fig. 7 (right), computed from the He-3 tables.

Insight: what "validated" means here

No constant in the solver was adjusted to match the paper. The tables, the energy balance and the HEM flux produce the paper's optimum, its maximum pressure, its mass flow and its cooling power to within about 1 %, and its fitted correlations to within a few per cent. The consistency lines show that $q_C = \min(\Delta h_T, q_L)$ and the optimum condition $\Delta h_T = q_L$ emerge from the exchanger balance to round-off — the paper's analytical results, re-derived numerically.

9.3 Things the reproduction taught

The optimum in T_{pre} is a plateau edge

At $P_U = 0.358$ MPa, q_C stays at 7.98 kJ/kg for every T_{pre3} below 8 K and falls above it (Q_C drops from 18.6 mW at 8 K to 13.1 mW at 10 K). Δh_T itself peaks near 4 K and is far above q_L there — but q_C cannot exceed q_L. So the paper's 8 K is the *warmest* temperature that still delivers the full latent limit (Part II, Fig. 16.2).

q_C and Q_C have different optimum pressures

The paper notes (Sect. 5) that at $T_{\mathrm{pre3}} = 5$ K the q_C optimum is at 0.21 MPa but the Q_C optimum at 0.36 MPa, because the orifice flux keeps rising with P_U. Our solver gives 0.213 and 0.352 MPa. Figure 9.2 shows why: colder precooling reaches a common envelope at lower P_U, and the envelope peaks near 0.36 MPa at about 18.6 mW whatever the precooling, as long as it is cold enough.

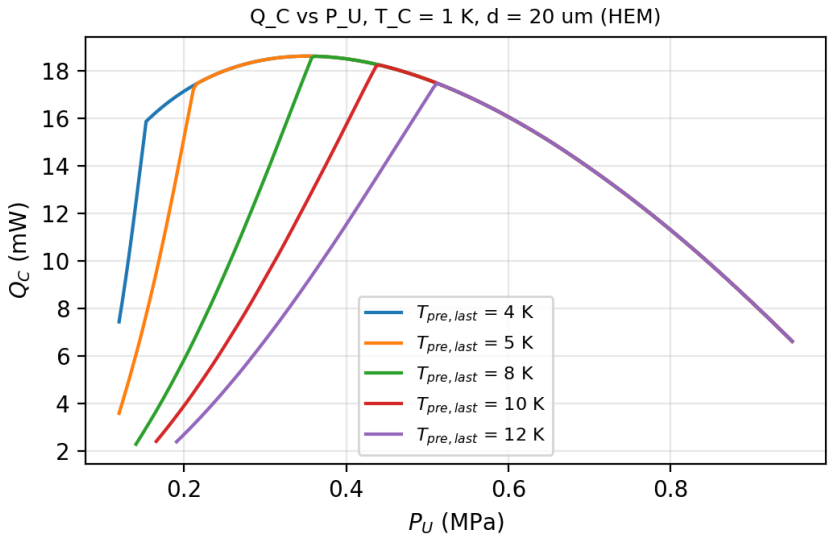


Figure 9.2: Q_C against P_U for several $T_{\text{pre,last}}$ (He-3, 1 K, 20 µm, HEM).

Insight: what to optimise when everything is free

If both P_U and $T_{\text{pre,last}}$ are free, q_C alone has no interior optimum: it keeps growing toward low P_U and cold T_{pre} while the mass flow goes to zero. The meaningful objective is $Q_C = \dot{m} q_C$. And since Q_C is nearly flat along the ridge $\Delta h_T = q_L$, a "practical" optimum — the warmest $T_{\text{pre,last}}$ and lowest P_U within 0.5 % of the maximum — is offered as the cheaper choice. The user picks which drives the design (`--drive Q_C|Q_C_practical|q_C`).

Why η_4 matters so much

η_4	T_8 [K]	$T_7 - T_{11}$ [K]	q_C [kJ/kg]	$\dot{m}$ [mg/s]	Q_C [mW]
1	1.000	0	7.961	2.339	18.62
0.99	1.349	0.078	7.475	2.322	17.36
0.97	1.910	0.219	6.503	2.267	14.74
0.95	2.333	0.360	5.531	2.197	12.15
0.90	3.086	0.712	3.102	1.994	6.19

Insight: the $(1-\eta)$ penalty multiplies a big number

From $q_C = \eta \min(\Delta h_T, q_L) - (1-\eta)(h_7 - h_{10})$: the penalty is $(1-\eta)$ times the enthalpy difference between the 8 K supply gas and the 1 K return vapour, $h_7 - h_{10} \approx 40.7$ kJ/kg — five times q_C itself. So 0.01 of effectiveness costs about 0.49 kJ/kg, i.e. 6 % of q_C. The flux falls only a little (3 % at 0.97) because T_8 rises. Together: about 7 % of Q_C per 0.01 of η_4, which is the paper's statement. The paper's design rule — cold-end difference below 1 K and warm-end difference below 0.2 K for $\eta_4 \geq 0.97$ — is met: 0.91 K and 0.22 K.

The flux at 1 K is not choked

At 1 K and 0.36 MPa the gas before the valve is a compressed liquid. Along its isentrope, $G(P)$ is largest at the back pressure: in HEM terms the flow is not choked, and the code says so (`UNCHOKED_RESTRICTION_FLOW`). The flux is still the HEM maximum and still matches the paper's

Table 2. It is a reminder that the “nozzle” picture of handwritten pages 7–8 is a model of a liquid squeezing through a 20 µm hole.

9.4 About the tolerances

The scorecard tolerances (`TOL` in `jt_run.py`) were set before the results were known and are loose: 15 % on a flux that agrees to 1 %, ± 0.03 MPa on a pressure that agrees to 0.002. A regression could pass unnoticed. Tightening them to about 2–3 % was offered and is still open.

Chapter 10

Designing with the solver

Once the paper was reproduced the question turned around: not "can we get the paper's numbers?" but "given only a target, what would we build?" This is the part the paper leaves open — it never says what T_{pre1} and T_{pre2} are — and it is what the handwritten objective asked for.

10.1 The discovery that simplified everything

Insight: the upstream precoolers cannot change the cooling

The last CHEX sees only two inlet states: the supply gas at $(P_U, T_{\text{pre,last}})$ and the saturated vapour at T_C. Its duty, T_8, q_C, and (through T_8) the orifice flux and $\dot{m}$ are fixed by those alone. So *no choice of the warmer precooler temperatures can change* q_C, $\dot{m}$ *or* Q_C. They only decide how the precooling load is shared between the stages.
This was confirmed numerically: sweeping T_{pre1} from 12 K to 300 K with everything else fixed leaves the total specific precooling at 11.468 kJ/kg (26.8 mW) and Q_C at 18.62 mW, while the share of PHEX 1 goes from 7.8 to 0 kJ/kg and PHEX 2 takes up the difference (Fig. 10.1). The global balance (paper's Eq. 22) says why the total is fixed: it is q_C plus the isothermal JT difference at 300 K.

This splits the design into two independent problems: (1) choose P_U and $T_{\text{pre,last}}$ for the refrigeration, (2) choose the upstream temperatures for the precooling. It also explains why the CHEX 1 pinch flag (Chapter 6) does not disturb the reproduction.

10.2 Choosing the upstream temperatures

Since the cooling does not care, the cryocooler should decide. Several ways of choosing were discussed; the one implemented (called criterion (b) in the code) minimises the ideal work needed to lift each precooler's heat to room temperature:

$$\Phi = \sum_j Q_{\text{pre},j}\left(\frac{T_H}{T_j} - 1\right), \qquad T_H > T_{\text{pre1}} > \cdots > T_{\text{pre,last}}, \quad Q_{\text{pre},j} \geq 0.$$

Heat removed at 10 K costs about 29 units of work per unit of heat; at 80 K, under 3. The constraint $Q_{\text{pre},j} \geq 0$ forbids a "precooler" that would have to *heat* the gas. Φ has kinks where a load reaches zero, so the search is differential evolution rather than a gradient method.

Decision: give a map, not just an answer

The Carnot weighting is a reasonable default but real cryocoolers are not Carnot machines, and their stages have fixed capacity curves. So for each free upstream temperature the code also scans that temperature alone (41 points) and plots every PHEX load against it. The designer reads the map against the capacity of a real cooler. This is what replaced the dropped "check against capacity curves" idea: the map states the requirement, the choice of cooler stays with the person.

For the paper's cooler with $T_{\text{pre,last}} = 8$ K the optimiser recovers $P_U = 0.3584$ MPa by itself, and places the upstream levels at 25.6 K and 12.3 K, with loads of 9.90, 8.32 and 8.56 mW. With the

paper-like placeholders (80, 25, 8 K) the loads would instead be 0.51, 9.70 and 16.61 mW: the same 26.8 mW total, but the 8 K stage carrying twice as much.

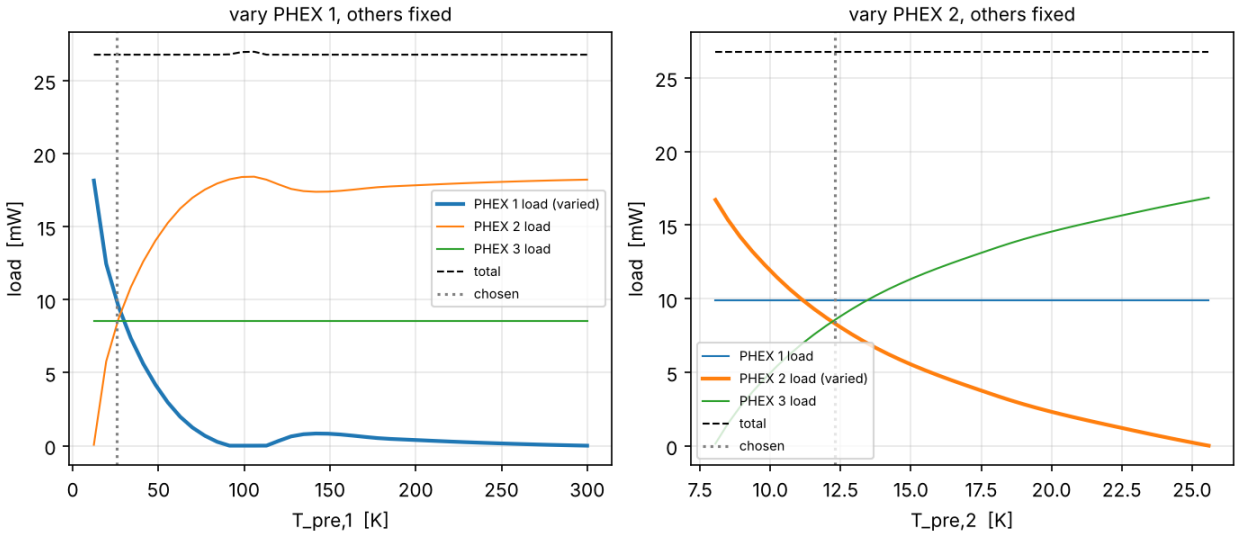


Figure 10.1: The upstream map for the paper's cooler: loads of the three PHEXs when the PHEX 1 (left) or PHEX 2 (right) temperature is swept. The total (dashed) does not move.

10.3 The feasibility check

A design tool that returns a number for an impossible design is worse than useless. So before solving, `check_config` asks, in order of cost, and answers in words:

1. Is T_C in the fluid's window? (He-3 0.5–3.2 K; CoolProp fluids triple point to 0.995 T_{crit}.)
2. Is $P_U > P_D = P_{\text{sat}}(T_C)$?
3. Do the given T_{pre} decrease strictly from T_H and stay above T_C?
4. Is $q_L = h_g(T_C) - h(P_U, T_C) > 0$? If not, no precooling can help; the message gives $P_{U,\text{max}}$.
5. Is $\Delta h_T > 0$ at $T_{\text{pre,last}}$? If not, the gas is above its inversion temperature and throttling heats it.
6. Is $T_{\text{pre,last}}$ warmer than $T_{\text{pre,opt}}$? Then q_C is limited by Δh_T — a warning, with the better value.
7. A trial solve, with its flags sorted into errors (negative PHEX load, no net refrigeration, energy balance failure) and warnings (pinch, unchoked flow).

Each step is the physics of an earlier chapter turned into a question. Designs it refuses, with the reason it gives: He-3 at $P_U = 1.2$ MPa ($q_L \leq 0$; $P_{U,\text{max}} \approx 1.13$ MPa); $T_{\text{pre}} = (80, 90, 8)$ K (not decreasing); $T_C = 3.4$ K (outside the tables); $T_{\text{pre,last}} = 60$ K (above inversion); nitrogen at $P_U = 0.1$ MPa (below $P_D = 0.137$ MPa); neon at 2 MPa (below its critical pressure, so the supply condenses inside a CHEX — a limitation of the model rather than of the design).

10.4 Trying other fluids and targets

Because the property layer is an interface, the same commands run He-4, neon, hydrogen, nitrogen and others through CoolProp. Some examples worked through during development:

Cooler	q_C [kJ/kg]	$\dot{m}$ [mg/s]	Q_C [mW]
He-3, 1.5 K, 3 CHEX, $T_{\mathrm{pre}} = (60, 10)$ K, 0.5 MPa, 15 µm	7.89	1.50	11.8
same with $T_{\mathrm{pre,last}} = T_{\mathrm{pre,opt}} = 9.39$ K	8.63	—	13.1
He-3, 2.0 K, 4 CHEX, $(80, 25, 10)$ K, 0.4 MPa, $\eta_4 = 0.99$, 20 µm	5.52	1.93	10.7
He-4, 4.2 K, 3 CHEX, $(80, 15)$ K, 0.8 MPa	6.21	2.90	18.0
Nitrogen, 80 K, 2 CHEX, 150 K, 15 MPa	117.8	41.6	4901
Hydrogen, 20 K, 3 CHEX, $(100, 40)$ K, 2 MPa	184.6	2.39	441

The second line shows the feasibility warning paying off: moving the last precooler from 10 K to the 9.39 K it suggested raised Q_C by 11 %.

10.5 A browser app that came and went

For a while the workflow lived in a local web page: a configuration form with the fluid, T_C, P_U (“blank = optimise”), T_H, orifice, number of CHEXs, one η and one T_{pre} per stage, optimiser settings (pressure and temperature ranges, flow model, “optimum that drives the cycle”, HEM grid points) and a data-reference block; and buttons to solve, optimise and save.

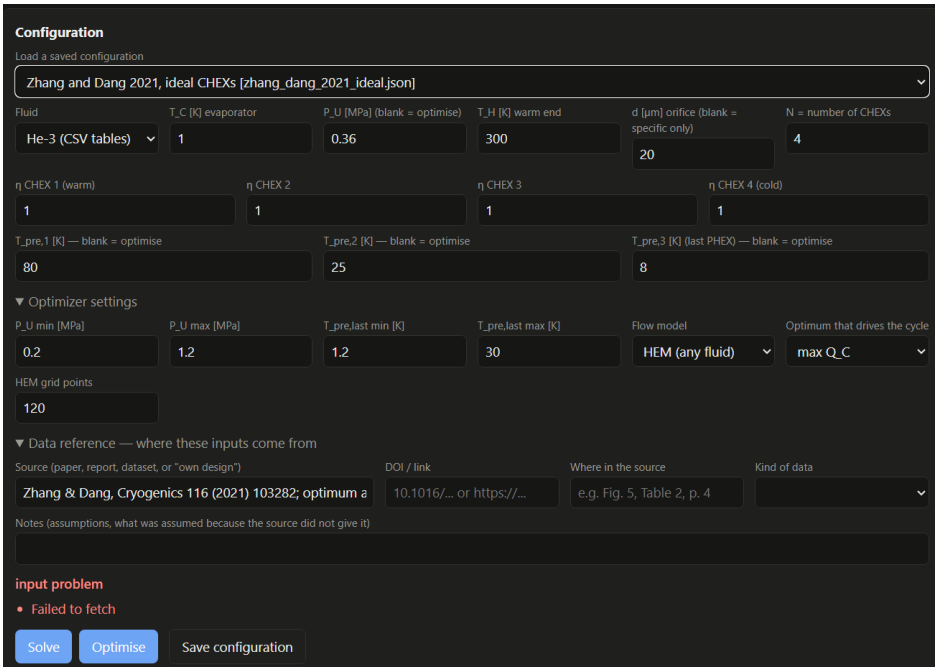


Figure 10.2: The browser front end, shortly before it was removed. The “Failed to fetch” message was a connection problem between the page and the local server.

In the conversation

Let’s forget about the interface, is everything else perfect? … Let’s completely remove the app thing. Let’s only have the script which could do all.

The form’s *structure* survived as the case-file format and the options of `jt_run.py run`; “blank = optimise” became writing `-` for an input.

10.6 Names, runs and references

Three bookkeeping decisions make results traceable:

- **Case names** say what the cooler is: `<Fluid>_<T_C>K_<d>um_<N>_stage`, plus `_eta<x>` when the last CHEX is not ideal, `<mdot>mgs` instead of the diameter for a measured flow, `nod` if neither, `Helium_4` for He-4, and `_v2`, `_v3` if the name exists with different content. (The scheme began with the phase-1 optimiser's output names, e.g. `Helium_3_2.1K_20um_4_stage`, and was extended.)
- **Every run is a folder**, `results/runs/<config>/<stamp>_<solve|optimize>/`, holding a copy of the configuration, a hash of its inputs, the results, the states and the plots. `jt_run.py runs` lists runs against configurations.
- **A data reference** (citation, DOI, where in the source, kind of data, notes on what was assumed) travels with each case and each run. It is deliberately *excluded* from the hash: adding or correcting a reference later must not orphan the runs made with that configuration.

Chapter 11

Other people's coolers

In the conversation

These are papers who use the same architecture.

Five papers were attached. Two are cited by Zhang & Dang: Liu et al. 2017 (their ref. 19) and Ma et al. 2019 in *Cryogenics* (ref. 20). The other three are the same Chinese group's 2.65 K cooler (Ma et al. 2019, *Sci. China*), the NIST 1.7 K cooler for nanowire detectors (Kotsubo, Ullom & Nam) and a closed-cycle He-3 JT cryocooler for thermometer calibration down to 0.65 K (Shimazaki et al., *Int. J. Thermophys.* 2008). All are a pulse-tube or GM cooler precooling a JT loop with counterflow exchangers — the same architecture, mostly with He-4.

11.1 What changed in the solver to read them

Decision: accept a measured mass flow

These papers report the mass flow they measured, not an orifice diameter — and their restrictions are multi-orifice elements, series orifices or capillaries that one diameter cannot describe. So the case format accepts `mdot_mg_s` *instead of* `d_um`. Giving both is rejected as over-specified. (The very first core had removed user-supplied flow on purpose; it came back for this reason.)

Decision: derive effectiveness from the paper's temperatures, never fit it

When a paper reports the temperatures at the ends of an exchanger rather than its effectiveness, η follows from the same definition as in the solver: for a warm-limited exchanger $\eta = (h(P_D, T_D) - h_C)/(h(P_D, T_A) - h_C)$. No parameter is adjusted to make a result match.

Each case is a JSON file in `cases/` with its inputs, a `reference` block saying where in the paper every number came from and what had to be assumed, and the paper's own values under `reported`. `python run_all.py` solves them all and scores the ones that are model-to-model comparisons.

11.2 Liu et al. (2017): the clean test

An open-cycle He-4 JT cooler; the paper gives $T_a = 17.3$ K entering the last exchanger, 1.137/0.104 MPa, $\dot{m} = 9.1$ mg/s, a *calculated* $Q_{\max} = 52.5$ mW and a measured 48.9–49.4 mW. $T_C = T_{\text{sat}}(0.104\ \text{MPa}) = 4.2517$ K. Their Eq. (1) has the return gas leaving the last exchanger at 17.1 K for supply entering at 17.3 K.

What went wrong: the first effectiveness was wrong

With $\eta_3 = 1$ the solver gave about 19 % too much. The effectiveness derived from 17.1/17.3 K was first written into the case note as 0.9971; recomputed properly from enthalpies it is 0.9857. With that, $q_C = 5.835$ kJ/kg (paper 5.77) and $Q_C = 53.1$ mW against 52.5 mW: +1.1 %. Lesson: a 0.2 K difference at 17 K looks like "almost ideal", but η is defined on enthalpies and the result is very sensitive to it. Derive it; do not eyeball it.

The 3.6 mW between Liu's calculation and measurement is heat leak, which the model does not have.

11.3 Ma et al. (2019), *Cryogenics* 104: three cases

A 100 mW at 5 K space JT cooler: precooling at 100 and 15 K, 2.0 MPa, a four-orifice element (4×20 µm). The paper bounds the exchanger effectiveness ("not lower than 0.97") without giving it, so $\eta = 0.97$ is an assumption.

- **Design point** (4.5 K, 10.5 mg/s, 100 mW design). Solver 90.2 mW at $\eta = 0.97$, 108.8 mW at $\eta = 1$; the design value corresponds to $\eta \approx 0.985$, inside the paper's bound. Precooler loads 207 and 278 mW against the design table's 350 and 260 mW: the second agrees within 7 %, the first does not. The upstream load depends on a precooler temperature the paper does not give for that comparison and on whatever parasitic loads its design figure includes, which the model cannot see.
- **Open loop, measured** (4.55 K, 11.08 mg/s, 100.3 mW). Solver 92.4 mW (-7.9 %), using design precooling temperatures because the test values are not given.
- **Closed loop, measured** (5.02 K, 11.02 mg/s, 102.3 mW). Solver 47.2 mW. Sensitivity runs traced the miss: at 5.02 K the model's evaporator pressure is $P_{\mathrm{sat}} = 0.199$ MPa, close to the He-4 critical pressure, where q_L is small; the paper quotes 0.16 MPa at the compressor suction. With $T_C = T_{\mathrm{sat}}(0.16\ \mathrm{MPa}) = 4.745$ K the solver gives 78 mW; with $\eta = 1$ at 5.02 K, 66 mW. The missing piece is the return-line pressure drop, i.e. the assumption $P_D = P_{\mathrm{sat}}(T_C)$ settled in the planning stage.

11.4 Ma et al. (2019), *Sci. China*: a 2.65 K cooler

Precooling 64.5/14.6 K, 0.6 MPa, $\eta = 0.972/0.973/0.972$ (given!), 1.2 mg/s, measured 1.48 mW at 2.71 K against a design of 4 mW. Solver: $q_C = 3.75$ kJ/kg, $Q_C = 4.50$ mW, $P_D = 15.1$ kPa (their return pressure: 12 kPa at the compressor, about 13.3 kPa at the evaporator). The solver lands on their *design* value; the authors themselves attribute the measured shortfall to lower mass flow, higher suction pressure and heat leakage. At 1.48 mW, about 3 mW of parasitic load is the whole story.

11.5 The two that could not be used

Kotsubo et al. give $P_U = 0.2$ MPa, $P_L = 3.2$ kPa, 0.6 mg/s, precoolers at 80, 25 and 10 K and 1.4 mW at 1.7 K, but no exchanger effectiveness. Shimazaki et al. give 1 mW at 0.65 K with 160 µmol/s (about 0.48 mg/s, so $q_C \approx 2.1$ kJ/kg) but neither P_U nor effectiveness. Given how strongly Q_C depends on η, a case built on guessed values would test the guess, not the solver; they were left as consistency checks only. Two further papers cited by Zhang & Dang (Narasaki, refs. 25 and 29) were never obtained.

11.6 What the comparison says

Case	Solver [mW]	Paper [mW]	
Liu 2017 (calculated)	53.1	52.5	+1.1 %
Ma 2019 design	90.2	100	−9.8 %
Ma 2019 open (measured)	92.4	100.3	−7.9 %
Ma 2019 closed (measured)	47.2	102.3	−54 %
Ma 2019 Sci China (measured)	4.50	1.48	+204 %

Insight: compare like with like

Where a paper reports an *ideal-model* number made under the same assumptions (Zhang & Dang; Liu's calculated Q_{max}), the solver agrees to about 1 %. Where it reports a *measurement*, the solver gives the ideal upper bound, and the gap is the heat leak and pressure drop it leaves out — and each gap was traced to a specific missing effect by a sensitivity run, not explained away. In `run_all.py` only the model-to-model comparisons are pass/fail (Liu Q_C 3 %, P_D 1 %; Ma design Q_C 12 %, P_D 1 %, second precooler 10 %); the measured ones are printed for information.

Chapter 12

Making it usable

12.1 One front door

After the app was removed, everything went through one script, `jt_run.py`, with a command per job:

Command	Job
`check`	are the He-3 tables present and sane?
`validate [-quick]`	data check, unit tests, paper reproduction, scorecard
`new -fluid ...`	create a named case with a data reference and check it
`feasible case.json`	can this design work, and if not, why?
`run case.json`	solve; or optimise whatever is left blank (`-`)
`runs`	list saved runs and their configurations
`solve cases/*.json`	solve cases and compare with the papers' reported values

and one script to run everything in order, `run_all.py` (`run_all.bat` on Windows): data check, paper 1 scorecard, then every other-paper case with a summary in `results/validation_all/`. A small bug remembers the Windows/PowerShell habit of writing blanks with a dash: `--T_pre -,-,8` was read by the argument parser as an option; values after `--T_pre` and `--P_U` are now glued back on.

12.2 Checking on my own machine

In the conversation

What do I send you for you to validate the results? … Here.

The terminal output of `python jt_run.py validate --quick` and every case, run on my laptop (Windows, Python 3.14) from the course folder, matched the development numbers exactly: 27/27, the same SHA-256 for the tables, the same Q_C for every case. That is the point at which the solver became *mine* rather than something that worked somewhere else.

12.3 Tests

29 unit tests (`python -m unittest discover -s tests -t .`) now cover the cycle core and CHEX logic, the interpolator against CoolProp (with assertions this time), the scorecard, and the workflow layer: naming and versioning, blank temperatures, feasibility messages, run linking, measured flow (including that d and $\dot{m}$ together are rejected).

12.4 Explaining it

The way the work was written up changed several times, and each change says something about what was needed.

1. A theory-and-code PDF and a README: complete, but organised like a manual.
2. For the seminar, a storyline: *theory and assumptions of the paper → plan for a solver → theory to solver → validation of Zhang & Dang → other papers with the same architecture → using the solver to design a coupled hybrid JT cooler, mapping the upper-stage precooler temperatures and loads to select a cryocooler.*
3. A slide deck was started and rejected ("could you not build a slide deck but rather a document explaining all in detail"); a shared write-up followed and was judged too thin ("please be detailed man, why leave any details if we spent so much time discussing the details"); then a paper-style document meant to be read beside Zhang & Dang.
4. Finally this journal, because "a lot of discussion and insights that are necessary from the chat, and even from the handwritten, are just not there".

Insight: results are not the whole of the work

The paper-style write-up had every number and still missed the point: the seminar expects — and future-me needs — the reasoning. Why the effectiveness is defined the way it is, why upstream temperatures do not matter, why the measured coolers disagree: these are as much results as 18.62 mW.

Chapter 13

The final stretch: deciding what counts, and showing it

By this point the solver was finished. What remained was to decide what the comparison should actually claim, to regenerate everything on fine grids, and to present it. Each of these changed the project more than I expected.

13.1 Comparing only with ideal models

In the conversation

I don't want to include the experimental data from other papers cause 1. too far, not fitting at all 2. pointless since our aim is to see calculations.

Decision: compare the ideal model only with ideal-model numbers

The measured cases (Ma's open and closed loops, the *Sci. China* cooler) never matched, and they never could: a measurement contains heat leak and return-line pressure drop, which the model leaves out on purpose (A2, A3). Comparing with them tests the assumptions, not the implementation. So the validation keeps only numbers that the source authors computed under the same assumptions: Liu's calculated $Q_{\max}$ (53.1 vs 52.5 mW, +1.1 %) and Ma's 100 mW design point (bracketed by 90.2–108.8 mW over the admissible $\eta = 0.97$–1, i.e. $\eta \approx 0.985$; 15 K load 278 vs 260 mW). The measured case files stay in `cases/` as informative only.

Insight

This is the cleanest statement of what the solver is: an exact implementation of an ideal model. "Agrees with ideal calculations to about 1 %" is a claim it can defend. "Predicts a real cooler" is not, and was never the aim.

13.2 Paper figures beside ours, not redrawn

In the conversation

I needed direct screenshots from the paper, not recreations.

What went wrong: recreating the paper's figures

My first comparison figures redrew the paper's curves from its fits, and a temporary change to `cycle_core` was made to get there. Both were wrong in principle: a recreated paper figure is our interpretation of the paper, so placing it next to our output compares us with ourselves. The change to the core was reverted and those scripts deleted.

The figures were instead extracted directly from the PDF (`pdfimages`) and placed side by side with the solver's own plots for identical inputs: Figs. 5, 7, 11, 13, 15 and 16. Optimum markers were changed from stars to open circles.

13.3 Dense grids

The coarse validation grid was good enough for pass/fail but made the curves angular and the statistics grid-dependent. `reproduce_zhang_dang_2021.py` gained `--dense` (241-point curves, 61-point optimisation grid, 40 precooling temperatures, 46 flux points, η step 0.0025) and individual overrides; `make_extra_figures.py` produces the Q_C–P_U family and the He-3 c_p curve. The latter appeared to hang on a long run, so it now prints a progress bar with elapsed time and an estimate of the time left. The heavy runs were done on my own machine. Two small command-line lessons: `new --T_pre -,-,8` fails because `argparse` reads `-,-,8` as an option, so the form is `--T_pre=-,-,8 --P_U=-`.

Dense results, which now replace the earlier ones everywhere:

Quantity	Dense value
$P_{U,\mathrm{opt}}$ vs Eq. 36, mean / worst	1.39 % / 3.72 %
$q_{C,\mathrm{opt}}$ vs Eqs. 35–36, mean / worst	3.12 % / 10.4 % (at 30 K)
q_C fit at our own $P_{U,\mathrm{opt}}$, mean / worst	0.17 % / 0.47 %
HEM flux vs Table 2, mean, $T_C = 1/2/2.6/3$ K	0.25 / 0.41 / 0.51 / 0.69 %
HEM flux vs Table 2, worst	0.71 / 0.72 / 2.28 / 1.49 %
Eq. 49 form refitted, worst	0.24 / 0.53 / 1.00 / 1.82 %
$T_{\mathrm{pre3,opt}}$ vs Eq. 36 at 0.3/0.5/0.7/0.9 MPa	0.01 / 0.23 / 0.34 / 0.78 K

The headline values ($P_{U,\mathrm{opt}}$, $q_{C,\mathrm{opt}}$, $\dot{m}$, Q_C) do not change with the grid; only the statistics over fitted ranges move, by a few tenths of a percent.

13.4 The presentation

The final presentation was 10 minutes (7 to talk, 3 for questions) by three presenters. The first version was too sparse; the second, an HTML deck, did not look academic. It was rebuilt in Beamer with typeset equations, the paper and solver figures side by side, each comparison table on the same slide as its figure, and speaker notes in a separate PDF. Several things were removed on request: the P–h diagram, part labels and presenter tags, "27/27" as a slogan, P_D (replaced by P_U), and the closing "thank you" line. The order: Vallabh — motivation, objectives, model, the last exchanger; Gaurav — solver structure and the full validation against the paper, ending with the summary of checks; Siddhant — other coolers, sizing the precooling stages, the Carnot example, lessons learned and conclusions.

Insight: the design result, stated once

The upper precooler temperatures cannot change q_C, $\dot{m}$ or Q_C; they only divide a fixed total load (26.8 mW for the paper's cooler). With the paper's $(80, 25, 8)$ K the split is $(0.5, 9.7, 16.6)$ mW, most of it at 8 K; the Carnot-weighted minimum, $(25.6, 12.3, 8)$ K, gives $(9.9, 8.3, 8.6)$ mW.

13.5 What came out at the end

Three documents: this journal (the story), a formal report (`report/jt_report.pdf`, everything here without the learning process), and the presentation. Part II of this journal is now the same text as the report.

Chapter 14

Looking back

14.1 From handwritten idea to code

In the handwritten derivation	In the solver
Assumptions: no pressure loss; T after PHEX $= T_{\text{pre}}$; He-3	Same, plus explicit: no heat leak, $C_d = 1$, $P_D = P_{\text{sat}}(T_C)$, any fluid
CHEX with A'; $\eta = (h_D - h_C)/(h_{A'} - h_C)$	$\eta = Q/\min(W_{\text{warm}}, W_{\text{cold}})$; the handwritten form is the warm-limited case
"coupled equations solved implicitly"	one explicit sweep from the evaporator upward
c_p argument for cold/hot-end pinch	enthalpy limits decide which end binds; 20-point interior scan for pinches inside
$q_C = \min(\Delta h_T, q_L)$ from qualities	emerges from the exchanger limits, checked to 10^{-14}
optimum at 8 K, 0.36 MPa (a maximum)	0.358 MPa; 8 K is the warmest point of a plateau; optimise Q_C when both are free
fits "to avoid calculation"	computed directly each time; the paper's fits used only as comparison
nozzle: $G = \Gamma G_{ID}$, throat by $V^* = a$, fit Γ	maximum of $G(P)$ along the isentrope (HEM); same choking condition
$Q_C = \frac{\pi d^2}{4}\Gamma G_{ID} q_C$, optimum over (T_{pre}, P_U)	$Q_C = \dot{m} q_C$ with $\dot{m}$ from HEM or measured; optimiser over $(P_U, T_{\text{pre,last}})$
$P_U, T_i, Q_{\text{pre},i}$ from T_C and Q_C	P_U and $T_{\text{pre,last}}$ from the optimiser; T_i by Carnot weighting plus a map; $Q_{\text{pre},i}$ for every stage

14.2 The insights worth keeping

1. **An exchanger has two limits.** Define effectiveness against the smaller one and the paper's minimum rule and optimum condition fall out of the energy balance.
2. **The optimum is where both ends of the last CHEX pinch**: $T_8 = T_C$ and the return gas leaves at T_{pre}.
3. **The best precooling temperature is the warmest one on a plateau**, not a peak.
4. **q_C and Q_C peak at different pressures**, because the orifice flux keeps growing with P_U.
5. **Upstream precoolers cannot change the cooling**; they only share a fixed total load. Choose them by what the cryocooler can do.
6. **The last CHEX is the critical component**: its $(1 - \eta)$ penalty multiplies an enthalpy gap five times the refrigeration.
7. **The choked flux is a maximum along an isentrope**; no speed of sound is needed.
8. **A formula will be evaluated outside its range** unless the physics that limits it is in the code.
9. **Compare ideal with ideal.** Measured coolers disagree with an ideal model by exactly the effects it leaves out; find which.
10. **Never fit a parameter to make a validation pass**; derive it from the source or leave the case out.

14.3 What is still open

- **Required $Q_C \rightarrow$ orifice diameter.** The handwritten objective designs from T_C *and* Q_C. The solver takes d (or $\dot{m}$) and returns Q_C; the inversion $d = \sqrt{4Q_C/(\pi q_C G)}$ is one line and has not been added.
- **Scorecard tolerances** are looser than the agreement obtained.
- **Heat leak and return-line pressure drop** would be the next physics to add if measured coolers are to be predicted.

Deliberately *not* pursued, after discussion: an inverse CHEX solver, the cooler-to-JT inverse problem, coupled checks against cryocooler capacity curves, geometry-based exchanger and physical PHEX or cryocooler-stage models.

14.4 Where to find things

Path (project folder)	Content
`jt/cycle_core.py`	states, CHEX limits and solver, pinch check, HEM, `solve_cycle`, CoolProp adapter
`jt/properties.py`	the He-3 interpreter
`jt/design_optimize.py`	last-stage optimiser, Eq. 49, naming
`jt/phex_load_map.py`	original PHEX load map
`jt/workflow.py`	configurations, feasibility, optimiser pipeline, upstream choice, runs
`jt_run.py, run_all.py`	the front door and the full validation chain
`reproduce_zhang_dang_2021.py`	the paper reproduction
`cases/`	case files with references
`data/he3/`	the He-3 tables
`results/`	everything generated (safe to delete)
`dev_history/`	the sources of this journal

Part II

Technical reference

About Part II

Part II is the same text as the formal report (`report/jt_report.pdf`): the compact, complete technical description of the solver and its validation, written to be read beside Zhang & Dang's paper: notation, the full derivations with every assumption, the property model, the implementation and theory-to-code map, the complete validation tables, the other papers with their inputs, the design application, limitations and reproducibility. Part I tells how each of these came about; Part II is where to look up the exact statement.

Chapter 15

System and notation

15.1 Architecture

Figure 15.1 shows the architecture used by the paper (Fig. 1 there) and by every case in this document, for N CHEXs and $N-1$ PHEXs; the paper has $N=4$. High-pressure gas at P_U enters the warm end of CHEX 1 at T_H (300 K). It leaves CHEX i, is cooled by PHEX i to $T_{\mathrm{pre},i}$, and enters CHEX $i+1$. After the last CHEX it is throttled through the orifice (isenthalpic) to the low pressure P_D, partly evaporates at T_C in the evaporator (where the refrigeration Q_C is absorbed), and the saturated vapour returns through all CHEXs, cooling the incoming stream, to leave at the warm end.

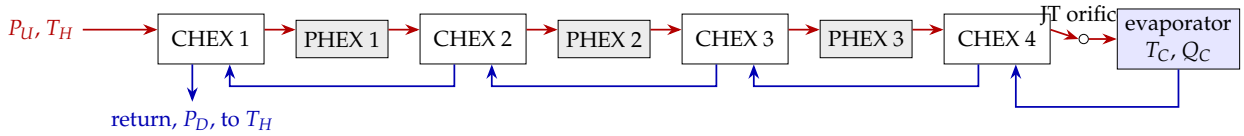


Figure 15.1: Architecture: red = high-pressure supply stream, blue = low-pressure return stream (schematic; cf. Fig. 1 of Zhang & Dang). PHEX j removes heat $Q_{\mathrm{pre},j}$ from the supply stream and is anchored to a stage of the precooling cryocooler.

15.2 State points

We follow the paper's numbering (its Fig. 2) for $N=4$. High-pressure side: 1 (CHEX 1 hot inlet, T_H), 2 (CHEX 1 hot outlet = PHEX 1 inlet), 3 (PHEX 1 outlet = CHEX 2 inlet, $T_{\mathrm{pre},1}$), 4, 5 ($T_{\mathrm{pre},2}$), 6, 7 ($T_{\mathrm{pre},3}$), 8 (CHEX 4 hot outlet, before the orifice). 9 is after the orifice (two-phase, $h_9 = h_8$), 10 is saturated vapour leaving the evaporator, 11, 12, 13 are the return-stream states between CHEXs and 14 is the warm-end exit. Points 1D, 3D, 5D, 7D are the states on the low-pressure isobar at the temperatures of points 1, 3, 5, 7. The code uses exactly this labelling for $N=4$ and generalises it for other N (Appendix A); Fig. 15.2 shows the resulting cycle in the P–h plane.

15.3 Symbols

h, s, ρ	specific enthalpy, entropy, density
P_U, P_D	high pressure; low pressure $= P_{\mathrm{sat}}(T_C)$
T_C, T_H	evaporator (cold-end) temperature; warm-end temperature (300 K)
$T_{\mathrm{pre},j}$	gas temperature leaving PHEX j (at P_U)
η_i	effectiveness of CHEX i (Section 16.3)
q, Q	per unit mass flow [kJ/kg], and power $Q = \dot{m}q$ [mW]
q_C, Q_C	refrigeration at the evaporator
$q_{\mathrm{pre},j}$, $Q_{\mathrm{pre},j}$	heat removed at PHEX j
Δh_T	isothermal JT enthalpy difference $h(P_D, T_{\mathrm{pre,last}}) - h(P_U, T_{\mathrm{pre,last}})$
q_L	latent limit $h_g(T_C) - h(P_U, T_C)$
G, d	orifice mass flux [kg m^{-2}s^{-1}]; orifice diameter

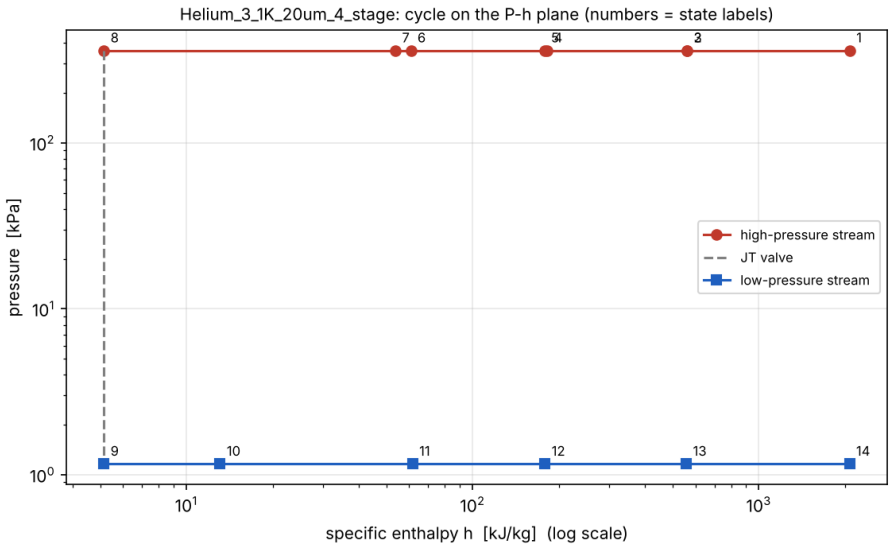


Figure 15.2: Cycle of the design point of the paper ($T_C = 1$ K, $P_U = 0.358$ MPa, $T_{\mathrm{pre}} = (25.6, 12.3, 8)$ K, $d = 20$ µm) on the P–h plane (log–log), with the state labels. Output of `jt_run.py run`; see Section 22.

Chapter 16

Theory: the physical model and its assumptions

16.1 Assumptions, stated once

All results below follow from the following assumptions. Items marked $*$ are the paper's; the others are choices we made explicit because they matter when comparing with other work.

A1* Steady state; one-dimensional counterflow; the two streams carry the same mass flow $\dot{m}$ (no leaks between them, no mass storage).

A2* No heat in-leak from the surroundings and no axial conduction along the exchangers. All heat entering the system enters at the evaporator (Q_C) or leaves at the PHEXs.

A3* No pressure drop along either stream: the high-pressure stream is at P_U everywhere, the low-pressure stream at P_D everywhere.

A4* The orifice is adiabatic: $h_9 = h_8$.

A5* The evaporator is at saturation: the stream leaving it is saturated vapour at T_C, and $P_D = P_{\mathrm{sat}}(T_C)$ (zero quality-independent temperature gradient in the evaporator).

A6* The precooler delivers the gas to a specified temperature $T_{\mathrm{pre},j}$; how the cryocooler achieves this (its own capacity curve) is outside the model.

A7 Each CHEX is characterised by one number, the effectiveness η_i, defined from an enthalpy balance, not by UA and length. Properties vary through the exchanger as the real fluid dictates (no ideal-gas shortcuts).

A8 The mass flow is set by the orifice (choked flow of the real fluid), or given from a measurement. It is never obtained from a compressor model.

A9 No discharge-coefficient correction ($C_d = 1$) and no multi-orifice effects.

Assumptions A2, A3 and A9 are optimistic and are why a real cooler delivers less than the model (Sections 21 and 23).

16.2 Energy balance of one counterflow exchanger

Label the four ends of a CHEX: hot stream in A (at P_U), hot stream out B; cold stream in C (at P_D), cold stream out D. With A1–A3 the first law for each stream gives the same heat duty Q per unit mass flow:

$$h_B = h_A - Q, \qquad h_D = h_C + Q. \tag{1}$$

Two physical limits bound Q. The *warm-end limit*: the cold stream cannot leave warmer than the hot stream enters, so $h_D \le h(P_D, T_A)$, i.e.

$$Q \le Q_{\mathrm{warm}} = h(P_D, T_A) - h_C. \tag{2}$$

The *cold-end limit*: the hot stream cannot leave colder than the cold stream enters, $h_B \ge h(P_U, T_C)$ with T_C now the temperature of state C:

$$Q \le Q_{\mathrm{cold}} = h_A - h(P_U, T_C). \tag{3}$$

$Q_{\max} = \min(Q_{\mathrm{warm}}, Q_{\mathrm{cold}})$. This is the enthalpy (not temperature) form of the effectiveness definition and it is essential for real gases: because the specific heat of the high-pressure stream differs strongly from that of the low-pressure stream (Fig. 16.1), the two limits are generally different and *which one binds is a result, not an assumption*. For He-3 at 0.36 MPa, c_p rises to about

12 kJ/kg/K near 5 K and falls to about 1.3 kJ/kg/K at 1 K, while at 1.16 kPa $c_p \approx 7$ kJ/kg/K throughout; the high-pressure stream has the larger capacity rate above ≈ 4 K and the smaller one below it, which is why the last CHEX is cold-end limited and the others are warm-end limited (this is the "pinch" discussion of the paper, its Fig. 4).

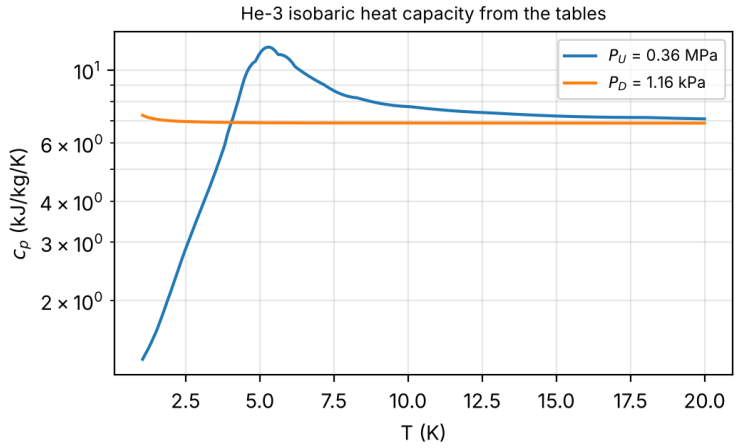


Figure 16.1: He-3 isobaric heat capacity from the tables at $P_U = 0.36$ MPa and $P_D = 1.16$ kPa (cf. Fig. 4 of the paper). Computed from `h(P,T)` by finite differences.

16.3 Effectiveness: the paper's definition and ours

The paper defines the effectiveness of the last CHEX from the cold side only (its Eq. 4):

$$\eta_4 = \frac{h_{11} - h_{10}}{h_{7D} - h_{10}}. \tag{4}$$

This is Q/Q_{warm}. We define instead

$$\eta = \frac{Q}{Q_{\max}}, \qquad Q_{\max} = \min(Q_{\text{warm}}, Q_{\text{cold}}). \tag{5}$$

The two coincide whenever the exchanger is warm-end limited, and for the paper's last CHEX at and below the optimal P_U (the regime of its analysis). When P_U is above the optimum the last CHEX is cold-end limited, $Q_{\text{cold}} < Q_{\text{warm}}$, and Eq. (4) with $\eta < 1$ would permit $Q > Q_{\text{cold}}$, i.e. $T_B < T_C$: the hot stream leaves colder than the cold stream enters, which is unphysical. Definition (5) is therefore used throughout, and we state this as a deliberate, checkable difference from the paper (it does not change any number in the paper's regime; see the validation in Section 20). The same definition makes η_i directly comparable with the effectiveness quoted by other authors from measured temperatures (Section 16.9).

16.4 The last stage: $q_C = \min(\Delta h_T, q_L)$

Consider the last CHEX, orifice and evaporator (states 7–11, paper Fig. 3). The refrigeration absorbed per unit mass is the enthalpy rise in the evaporator,

$$q_C = h_{10} - h_9 = h_{10} - h_8, \tag{6}$$

with $h_{10} = h_g(T_C)$ and $h_8 = h_7 - Q$ from Eq. (1). For an ideal CHEX ($\eta = 1$) $Q = Q_{\max}$ and Eq. (6) becomes

$$\text{warm-end limited: } q_C = h_{10} - h_7 + (h_{7D} - h_{10}) = h(P_D, T_7) - h(P_U, T_7) \equiv \Delta h_T, \tag{7}$$

$$\text{cold-end limited: } q_C = h_{10} - h(P_U, T_C) \equiv q_L. \tag{8}$$

Because the limiting case is the one with the smaller Q, which is the one giving the larger h_8 and hence the *smaller* q_C,

$$q_C = \min(\Delta h_T,\, q_L), \tag{9}$$

which is the paper's Eqs. (30)–(31). Δh_T is the isothermal JT enthalpy difference at the precooling temperature (how much enthalpy the gas gives up when throttled isothermally at T_7); q_L is the latent heat of the liquid at the evaporator temperature compressed to P_U (the most the liquid could ever absorb on evaporation).

Dependence on pressure. From $(\partial h/\partial P)_T = v(1 - \beta T)$ (β the volume expansivity),

$$\Delta h_T(P_U) = \int_{P_D}^{P_U} v\,(\beta T - 1)\,\mathrm{d}P, \qquad q_L(P_U) = h_g - h(P_U, T_C), \quad \frac{\mathrm{d}q_L}{\mathrm{d}P_U} = -v_L(1 - \beta_L T_C) < 0. \tag{10}$$

For the liquid $\beta_L T_C \ll 1$, so q_L falls almost linearly with P_U and vanishes at a maximum pressure $P_{U,\max}$ (1.131 MPa for He-3 at 1 K; paper: 1.13 MPa), above which no refrigeration is possible at any precooling temperature. Δh_T rises with P_U (below the inversion temperature) and saturates. Hence $q_C(P_U)$ increases with Δh_T until the two curves cross, then decreases along q_L: the maximum is at the crossing,

$$\Delta h_T(P_{U,\mathrm{opt}}, T_{\mathrm{pre}}) = q_L(P_{U,\mathrm{opt}}), \tag{11}$$

the paper's Eq. (34), and equality of the two is both necessary and sufficient for the optimum (Fig. 16.2, left). Equation (11) is one equation in two unknowns, so it defines a curve $P_{U,\mathrm{opt}}(T_{\mathrm{pre}})$ (paper's Fig. 8, fitted by its Eqs. 35–36).

Dependence on T_{pre}. Δh_T has a maximum in temperature (near 4 K for He-3 at 0.36 MPa, dashed green curve in Fig. 16.2, right) and falls strongly as T_{pre} rises; q_L does not depend on T_{pre}. While $\Delta h_T > q_L$ the refrigeration is on the plateau $q_C = q_L$: cooling the gas further does not help. The *warmest* T_{pre} on the plateau is therefore the best choice, $T_{\mathrm{pre,opt}}$: it delivers the same q_C with the least demanding precooler. At $P_U = 0.358$ MPa this is 8 K, which is the paper's choice of $T_{\mathrm{pre3}} = 8$ K.

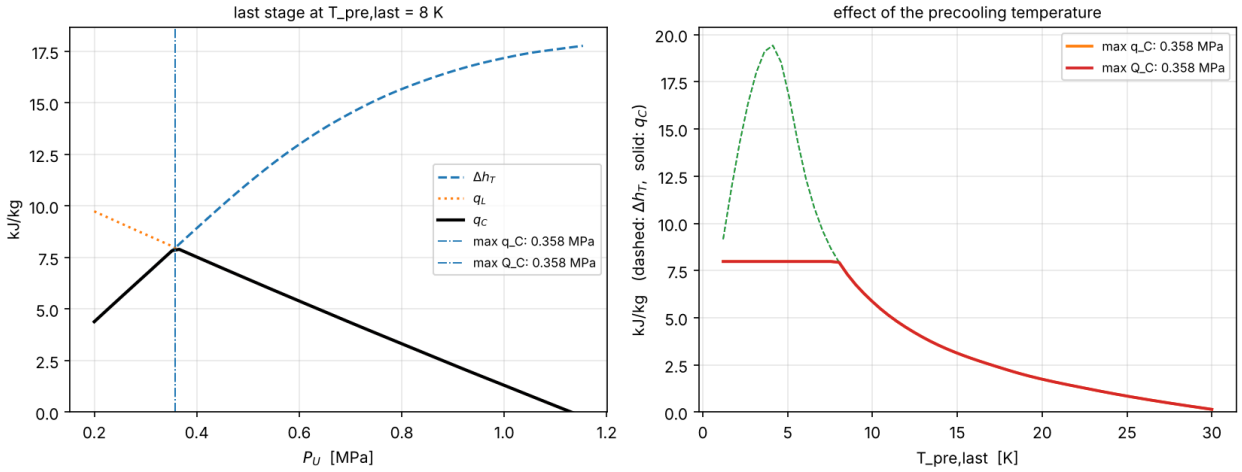


Figure 16.2: Last stage of the design example. Left: Δh_T, q_L and $q_C = \min$ versus P_U at $T_{\mathrm{pre,last}} = 8$ K; the optimum is the crossing at 0.358 MPa and the curve reaches zero at 1.13 MPa (cf. paper Fig. 5). Right: effect of $T_{\mathrm{pre,last}}$ (at the optimal P_U the plateau $q_C = 7.98$ kJ/kg ends at 8 K; cf. paper Fig. 7).

Imperfect last CHEX. For $\eta < 1$, $Q = \eta Q_{\max}$ and Eq. (6) gives

$$q_C = \eta\, Q_{\max} - (h_7 - h_{10}) = \eta\, \min(\Delta h_T, q_L) - (1 - \eta)\,(h_7 - h_{10}). \tag{12}$$

The penalty is $(1-\eta)$ times the *whole* enthalpy difference between the warm hot-stream end and the cold return stream (here $h_7 - h_{10} = 40.7$ kJ/kg), about five times the useful q_C. Hence a 1 % loss of effectiveness costs about 6 % of q_C (we find 7.96 $\rightarrow$ 7.47 kJ/kg for $\eta = 1 \rightarrow 0.99$), which is why the last CHEX is the critical component. Together with the associated mass-flow change (Section 16.6), this gives the large sensitivity of Q_C shown in Section 20.

16.5 Cascade balance and what the upstream precoolers can change

For ideal exchangers the paper obtains, from Eq. (1) applied to each CHEX and the precoolers (its Eqs. 18–22),

$$q_C = \sum_j q_{\mathrm{pre},j} + \left[h(P_D, T_H) - h(P_U, T_H)\right], \tag{13}$$

where the bracket is the (negative) isothermal JT enthalpy difference at ambient, and $Q_C = \dot{m} q_C$. Equation (13) is just the first law over the whole machine: everything the precoolers remove, plus the net enthalpy the compressed gas brings in, balances the refrigeration. For $\eta < 1$ the same balance holds with $h(P_D, T_{D1})$ in place of $h(P_D, T_H)$, T_{D1} being the actual warm-end return temperature; the solver evaluates it that way and checks it.

An important consequence. The last CHEX sees fixed boundary states on both streams: its hot inlet $(T_{\mathrm{pre,last}}, P_U)$ and its cold inlet (saturated vapour at T_C). Its duty, q_C, T_8, and through T_8 the orifice flux and $\dot{m}$, are therefore determined by T_C, P_U, $T_{\mathrm{pre,last}}$, η_{last} and the orifice alone. *The upstream precooler temperatures cannot change q_C, $\dot{m}$ or Q_C*; they only decide how the (fixed) total precooling load $\sum_j q_{\mathrm{pre},j}$ in Eq. (13) is distributed among the PHEXs. This separates the design problem into two stages: optimise the last stage (Section 16.7), then choose the upstream temperatures by a different criterion (Section 16.8). We verified the invariance numerically (Fig. 22.1: the total load and Q_C stay constant while any one upstream temperature is swept over its full range).

16.6 Mass flow: choked flow through the orifice

$Q_C = \dot{m} q_C$ needs $\dot{m} = \pi d^2 G/4$, with G the mass flux through the orifice. The paper computes G for real He-3 from the choked-flow condition using the fundamental derivative-based factor Γ and the real-gas equation of state (its Eqs. 41–48), and fits the results (Table 2) with an algebraic form (Eq. 49) as function of P_U and T_8. We use an equivalent but simpler formulation that needs only the property tables: a *homogeneous-equilibrium model* (HEM). For stagnation state (P_0, h_0, s_0) (state 8) and an isentropic expansion to pressure P,

$$G(P) = \rho(P, s_0)\sqrt{2\left[h_0 - h(P, s_0)\right]}, \qquad G_{\mathrm{choked}} = \max_{P_D \le P < P_U} G(P). \tag{14}$$

The maximum is found on 500 logarithmically spaced pressures and refined by golden-section search. If the maximum lies at $P = P_D$ the flow is not choked (flag `UNCHOKED_RESTRICTION_FLOW`); this is typical when the inlet is liquid-like. Equation (14) agrees with the paper's real-gas results to 0.25–0.72 % (mean) at $T_C = 1, 2, 2.6, 3$ K (Table 20.1). The paper's own fit (Eq. 49) is also implemented (`--flow-model paper_fit`) and is itself reproduced by our HEM data to 0.2–1.3 % in the worst case.

Why $\dot{m}$ depends on η. G depends on the stagnation state 8; with $\eta < 1$, T_8 rises (from 1 K to 1.91 K at $\eta = 0.97$), the fluid is less dense and its flux falls, hence the decrease of $\dot{m}$ with η in the paper's Fig. 15 (2.339 $\rightarrow$ 2.267 mg/s here).

16.7 Optimising Q_C rather than q_C

Since $Q_C = \dot{m} q_C$ and $\dot{m}$ increases with P_U (flux grows with upstream pressure), the pressure for maximum Q_C is above that for maximum q_C *when the flux gain outweighs the q_C loss*. In the paper's case with $T_{\mathrm{pre}} = 5$ K the q_C maximum is at 0.21 MPa but the Q_C maximum is at 0.36 MPa (paper, Sect. 5); we obtain 0.213 and 0.352 MPa (Section 20). A second subtlety is that q_C alone has *no* interior optimum in $T_{\mathrm{pre,last}}$ (it is monotone: colder is never worse), so "the optimum" only exists with a constraint or a different objective. We therefore (i) optimise Q_C when a flow is available, with a "practical" option that picks the warmest T_{pre} and the lowest P_U whose Q_C is within 0.5 % of the maximum (a flat optimum: Fig. 16.3), and (ii) use $T_{\mathrm{pre,opt}}$ of Eq. (11) when T_{pre} is to be chosen.

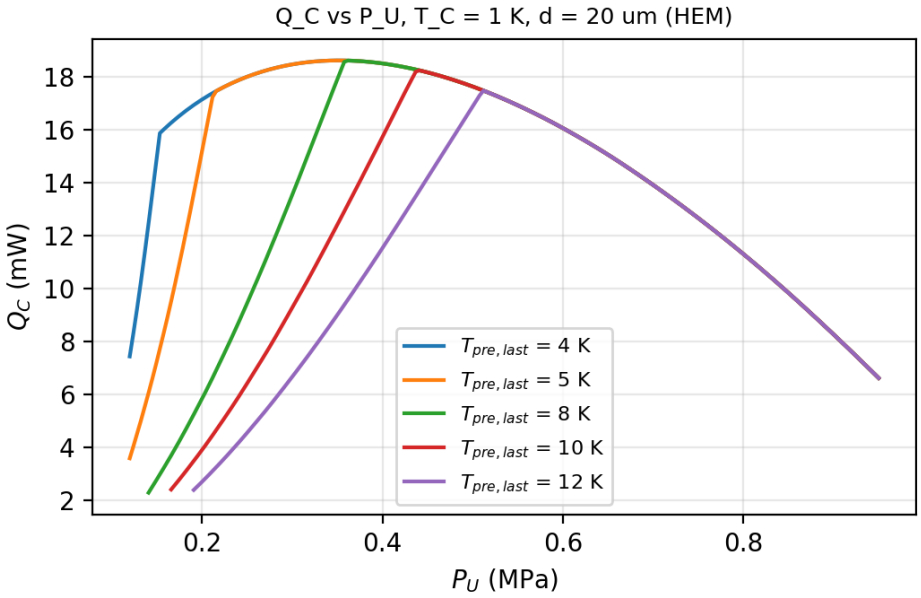


Figure 16.3: Q_C versus P_U at $T_C = 1$ K, $d = 20$ µm (HEM flux) for several $T_{\mathrm{pre,last}}$ (cf. Fig. 13 of the paper). Curves with colder T_{pre} reach the common envelope (maximum 18.6 mW at about 0.36 MPa) at lower P_U; warmer T_{pre} forces the optimum to higher P_U and lower Q_C.

16.8 Choosing the upstream temperatures

By Section 16.5 upstream temperatures $T_{\mathrm{pre},j}$ ($j < N-1$) leave q_C unchanged, so another criterion is needed. Physically, a precooler at temperature T_j rejecting Q_j to ambient T_H needs at least the Carnot work $Q_j(T_H/T_j - 1)$, and real cryocoolers are much more demanding at low T_j. We therefore minimise

$$\Phi = \sum_j Q_{\mathrm{pre},j}\left(\frac{T_H}{T_j} - 1\right) \tag{15}$$

over the upstream $T_{\mathrm{pre},j}$, subject to $T_H > T_1 > T_2 > \cdots > T_{\mathrm{last}}$ and every $Q_{\mathrm{pre},j} \geq 0$ (a PHEX may not heat the gas). A differential-evolution search is used, because Φ has non-smooth regions where a PHEX load reaches zero. This is only one defensible criterion (the real choice depends on the capacity curves of the cryocooler used, Section 22), so the code also produces the full map of loads versus each upstream temperature, from which the user can choose any other.

16.9 Prescribed flow and effectiveness from temperatures

Some papers give $\dot{m}$ and temperatures rather than an orifice diameter. The solver accepts a prescribed `mdot_mg_s` instead of d (d and $\dot{m}$ together are rejected as over-specified). If the paper reports the temperatures at the two ends of the last CHEX (e.g. hot stream in at T_A, return stream out at T_D), the effectiveness follows from Eq. (5) in the warm-limited case,

$$\eta = \frac{h(P_D, T_D) - h_C}{h(P_D, T_A) - h_C}. \tag{16}$$

This is used for the Liu et al. case (Section 21).

16.10 Internal pinch check

Eq. (1)–(5) constrain only the ends of an exchanger. With strongly varying c_p the stream temperatures could still cross *inside*. The solver therefore integrates the exchanger at 20 equal enthalpy steps, computing both stream temperatures, and flags a negative gap (`INTERNAL_TEMPERATURE_CROSSOVER`); this is a warning, not an error. In the paper's design CHEX 1 shows a small crossover (minimum gap -0.01 K): a sign that the ideal $\eta = 1$ is physically marginal in that exchanger, not a solver failure.

Chapter 17

Modelling decisions

The following decisions determine what the solver does and does not claim.

1. **Energy balance over UA models.** The paper works in the P–h plane with effectiveness; a UA–NTU model would need geometry and heat-transfer correlations that the target papers do not provide. All inputs are therefore thermodynamic: T_C, P_U, $T_{\mathrm{pre},j}$, η_i, d or $\dot{m}$.
2. **Properties from the paper's own tables.** For He-3 the supplied single-phase and saturation tables (not an equation of state) are the single source of truth; this removes the ambiguity of reference states and keeps the reproduction faithful. For other fluids CoolProp is used through the same interface.
3. **Everything generic in N.** The cycle is solved for any number of CHEXs, not only four, so other coolers (3 CHEX with 2 PHEXs, etc.) run unchanged.
4. **HEM for the orifice** (reasoned above), with the paper's own fit available for comparison.
5. **Fail loudly.** Inputs outside the tables raise an error (no extrapolation); the feasibility check explains infeasibility rather than returning a number.
6. **Validation first, claims second.** The solver was checked against all quantitative statements of the paper before it was applied to other papers; other-paper cases carry their reported values and a reference block, and mismatches are explained rather than tuned away (no free fitting parameter is used to force agreement).
7. **One front door.** A single command-line script (no GUI) drives validation, other papers and own designs; every run is stored with its configuration so that a result can always be traced to inputs.

Chapter 18

Property models

18.1 He-3 from the supplied tables

The paper's results depend on He-3 properties in a region (0.5–3.3 K, 1 Pa–3 MPa) where no standard equation of state package is available, so the solver reads two CSV tables (`data/he3/`): single-phase properties on a (T, P) grid (0.5–300 K, 1 Pa–3 MPa) and saturation properties (0.5–3.2 K). The loader checks columns, units, monotonicity and splits the saturation table into liquid and vapour branches; `python jt_run.py check` reports the SHA-256 of both files so a run can be tied to a data version (here `b17901e3e299` and `8a674a6a03da`) and confirms $P_{\mathrm{sat}}(1\,\mathrm{K}) = 1160.3$ Pa, which is the paper's 1.16 kPa.

18.2 Interpolation

- **Method.** $h(P,T)$, $s(P,T)$, $\rho(P,T)$ are interpolated with monotone piecewise-cubic Hermite polynomials (PCHIP), first along each tabulated isotherm in $\ln P$, then across isotherms in T at the requested pressure. PCHIP does not overshoot: crucial for $c_p = \partial h/\partial T$ where c_p has a sharp peak (Fig. 16.1).
- **Phase classes.** The table spans the liquid and vapour phases. For each (P,T) the phase is decided by comparing T with $T_{\mathrm{sat}}(P)$; interpolation is done only among nodes of the same class so that the discontinuity at saturation is not smoothed over. Above the critical point (3.3246 K, 116.4 kPa) a single class is used; the critical point is used for *classification*, closing the saturation dome.
- **Near-critical gap.** Between 101.7 kPa (last tabulated saturation point) and the critical pressure the tables have no saturation data; requests there raise `OutOfRange`. This is why the pressure sweeps of the Q_C study start at 0.12 MPa. A liquid-class lookup needs at least three nodes, hence it is not available below ≈ 45 Pa.
- **No extrapolation.** Anything outside the tables raises `OutOfRange`; T_C is limited to ≤ 3.2 K. This is deliberate (Section 17, item 5).
- **Inverse.** $T(P,h)$ is obtained by bracketed root finding on $h(P,T)$; a $T \to h \to T$ round trip at 0.36 MPa, 8 K returns 8.00000 K (part of the data check). Absolute values of h and s depend on the table's reference state; only differences enter any result, so the reference state is not checked.

18.3 Other fluids

For He-4, Ne, H_2, para-H_2, N_2, Ar, O_2, CH_4 and others, the same interface (`h, s, rho, Psat, T_from_Ph, ...`) is served by CoolProp. Two restrictions follow from the tables-free approach: the high pressure must exceed the critical pressure of the fluid (He-4 0.227 MPa is not an issue, but Ne 2.66, H_2 1.30, N_2 3.40 MPa are), otherwise the hot stream condenses inside the exchangers and the property lookup fails; and the feasibility check restricts T_C from the triple point to 0.995 T_{crit}.

Chapter 19

Implementation

19.1 Structure

File	Content
`jt/properties.py`	`Helium3Fluid`: CSV-backed tables, PCHIP interpolation, saturation, inverse, error classes
`jt/cycle_core.py`	state points, the CHEX solver (Eqs. 1–5), `solve_cycle`, the HEM choked flux (Eq. 14), the CoolProp adapter, flags
`jt/design_optimize.py`	last-stage optimisation of P_U, $T_{\mathrm{pre,last}}$; Eq. 49 fit; case-file naming
`jt/phex_load_map.py`	PHEX loads versus an upstream temperature
`jt/workflow.py`	configuration normalisation, feasibility check, optimiser pipeline, upstream search, run storage
`jt_run.py`	the command-line front door
`run_all.py`	one-command validation chain (data check, paper, other papers)
`reproduce_zhang_dang_2021.py`	reproduction of the paper's figures/fits
`tests/test_workflow.py`	29 unit tests

19.2 Solving a cycle (`solve_cycle`)

Given T_C, P_U, T_H, $T_{\mathrm{pre},1..N-1}$, $\eta_{1..N}$ and d or $\dot{m}$:

1. Compute $P_D = P_{\mathrm{sat}}(T_C)$ and the evaporator outlet $h_{10} = h_g(T_C)$.
2. **Last CHEX first.** Its hot inlet $(T_{\mathrm{pre,last}}, P_U)$ and cold inlet (h_{10}) are known; apply Eqs. (1)–(5) to get Q, hence h_8, h_{11} and the states on both streams. $q_C = h_{10} - h_8$.
3. **Work upstream.** The cold-stream inlet of CHEX i is the outlet of CHEX $i+1$ (known). The hot inlet of CHEX i is $(T_{\mathrm{pre},i-1}, P_U)$ (or T_H for $i = 1$). Apply Eqs. (1)–(5). The PHEX load follows from the enthalpies on either side: $q_{\mathrm{pre},i} = h(\text{CHEX } i \text{ hot out}) - h(P_U, T_{\mathrm{pre},i})$.
4. $\dot{m} = \pi d^2 G/4$ from Eq. (14) at the stagnation state 8 (or the given $\dot{m}$); $Q_C = \dot{m} q_C$, $Q_{\mathrm{pre},j} = \dot{m} q_{\mathrm{pre},j}$.
5. **Checks.** global balance Eq. (13), non-negative PHEX loads, temperature ordering, internal pinch scan, choking.

Because every step is explicit (no iteration on the cycle), a solve takes milliseconds, which makes the scans of Section 22 cheap; the cost is in the property lookups and the choked-flux search.

19.3 Theory-to-code map

Equation / statement	Paper	Where in the solver
CHEX energy balance, limits (1)–(3)	Eqs. 2, 5, 9–12	CHEX solver in `cycle_core.py`
Effectiveness (5) (differs from Eq. 4 beyond optimum)	Eq. 4	CHEX solver, `eta`
$q_C = \min(\Delta h_T, q_L)$ (9)	Eqs. 30–31	`solve_cycle` (emerges, not coded); tested by the scorecard check "$q_C = \min(\Delta h_T, q_L)$"
Optimum $\Delta h_T = q_L$ (11)	Eq. 34	`design_optimize.py` (scan, then root refinement)
Fits $P_{U,\mathrm{opt}}$, $q_{C,\mathrm{opt}}$	Eqs. 35–36, Table 1	compared in `reproduce_zhang_dang_2021.py`
Cascade balance (13)	Eq. 22	global-balance check in `solve_cycle`
HEM flux (14)	Eqs. 41–48	`cycle_core.py`
Paper's flux fit	Eq. 49, Table 2	`design_optimize.py`, `--flow-model paper_fit`
$Q_C = \dot{m} q_C$	Eq. 50	`solve_cycle`
Upstream criterion (15)	— (new)	`workflow.optimise_upstream`
Upstream sensitivity map	— (new)	`workflow.upstream_map`
Feasibility	— (new)	`workflow.check_config`

19.4 Feasibility check

`jt_run.py feasible case.json` answers "is this design physically possible and why not". In order: (1) T_C inside the fluid window (He-3 0.5–3.2 K; CoolProp fluids triple point to $0.995T_{\mathrm{crit}}$); (2) $P_U > P_D$; (3) T_{pre} strictly decreasing from T_H and all above T_C; (4) $q_L = h_g(T_C) - h(P_U, T_C) > 0$, otherwise no refrigeration at any T_{pre} (the message gives $P_{U,\max}$); (5) $\Delta h_T > 0$ at $T_{\mathrm{pre,last}}$, otherwise the gas is above its inversion temperature; (6) a warning if $T_{\mathrm{pre,last}} > T_{\mathrm{pre,opt}}$ (then $q_C = \Delta h_T$ is limited by the precooling); (7) a trial solve whose flags are classified as errors (a PHEX would have to heat the gas, balance fails, no net refrigeration) or warnings (internal pinch, unchoked flow). The verdict is *feasible*, *feasible with warnings* or *infeasible*.

19.5 Run storage, naming and the data reference

- **Case names.** `<Fluid>_<T_C>K_<d>um_<N>_stage[_eta<x>]`, e.g. `Helium_3_1K_20um_4_stage`; a prescribed flow replaces the orifice by `<mdot>mgs`, `nod` marks "no flow information", and `_v2`, `_v3` are appended if the name already exists with different content. He-4 is labelled `Helium_4`.
- **Run folders.** Every `run` writes `results/runs/<config name>/<stamp>_<solve|optimize>/` with `config.json`, `meta.json`, `result.json`, `states.csv`, `ph_cycle.png` and, for optimisation, `map_qC.png, map_QC.png, stage_curves.png, upstream_map.png/csv`.
- **Linking runs to configurations.** `meta.json` holds a hash of the input configuration (fluid, T_C, P_U, T_H, T_{pre}, η, d or $\dot{m}$, optimiser settings). The data-reference block (citation, DOI, location in the paper, kind, note) is *excluded* from the hash, so a reference can be added or corrected later without orphaning the runs. `python jt_run.py runs` lists the runs against configurations.

19.6 Command reference

Command	Purpose
`check`	He-3 tables present and consistent
`validate [-quick]`	workflow 1: data check, tests, paper reproduction, scorecard
`new -fluid ...`	create a named case with data reference, check feasibility
`feasible cases/x.json`	feasibility verdict and reasons
`run cases/x.json [-optimize ...]`	solve, or optimise if anything is blank
`runs`	list saved runs and their configurations
`solve cases/*.json`	plain solve with reported-value comparison and sweeps
`optimize, phex`	original last-stage optimiser and PHEX load map
`python run_all.py [-quick]`	everything in one command

Blank inputs (`-` for T_{pre} or P_U) trigger the optimiser; the options are `--drive Q_C|Q_C_practical|q_C`, `--flow-model hem|paper_fit`, `--pu-min`/`--pu-max`, `--tpre-min`/`--tpre-max`, `--hem-grid`, `--save-final`.

19.7 Where the generated files go

Location (relative to the project folder)	Content
`results/validation/`	paper reproduction report, scorecard (`scorecard.json`), figures
`results/validation_all/`	`run_all.py` summary (`summary.csv`, `summary.json`), per-case results
`results/cases/<name>/`	output of `solve` (result, states, sweeps)
`results/runs/<config>/<stamp>_<kind>/`	output of `run` (Section above)
`cases/`	case files (inputs, references, reported values)
`paper_figs/ (delivered with this document)`	the figures and data used here

`results/` is generated output and can be deleted at any time.

Chapter 20

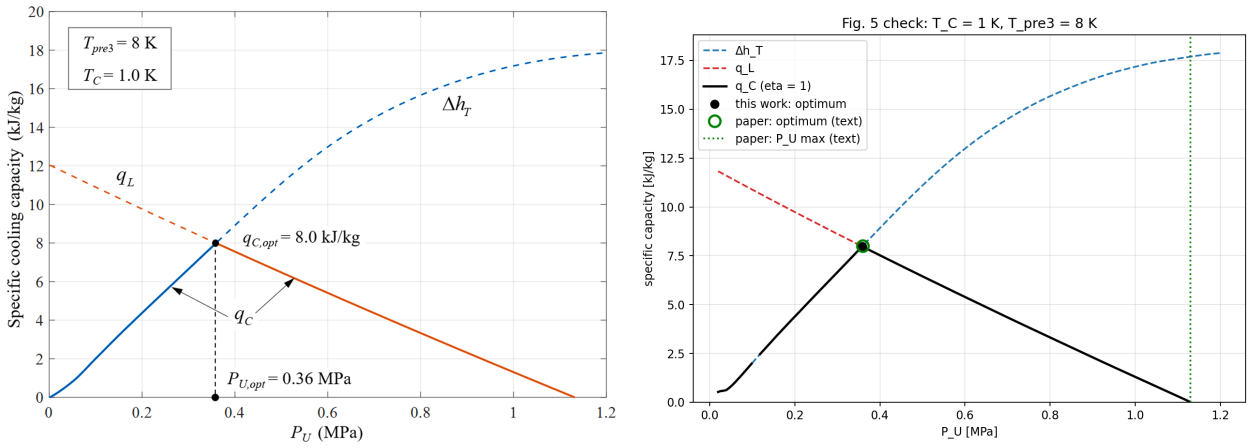
Validation against Zhang & Dang (2021)

All values are produced by `python jt_run.py validate --dense` (curves with 241 points, optimisation grids of 61 points, 40 precooling temperatures, 46 flux points, effectiveness step 0.0025). Inputs are those of the paper: He-3, $T_C = 1$ K, $T_H = 300$ K, $d = 20$ µm, $T_{\mathrm{pre}} = (80, 25, 8)$ K. In each comparison figure, panel (a) is extracted directly from the paper and panel (b) is the solver output for the same inputs; the paper's figures are not redrawn.

20.1 Last-stage optimum (paper Figs. 5–8)

Quantity	Solver	Paper	Difference
$P_{U,\mathrm{opt}}$ at $T_{\mathrm{pre3}} = 8$ K [MPa]	0.3584	0.36	−0.4 %
$q_{C,\mathrm{opt}}$ [kJ/kg]	7.979	8.0	−0.3 %
$P_{U,\max}$ ($q_L = 0$) [MPa]	1.131	1.13	+0.1 %
P_U for max q_C, $T_{\mathrm{pre}} = 5$ K [MPa]	0.213	0.21	+1.4 %
P_U for max Q_C, $T_{\mathrm{pre}} = 5$ K [MPa]	0.352	0.36	−2.2 %
$\Delta h_T = q_L$ at the optimum (Eq. 11), max. rel. gap	7.5×10^{-8}	—	identity
$q_C = \min(\Delta h_T, q_L)$, max. abs. deviation	4×10^{-14}	—	identity

The last two rows confirm that the minimum rule and the optimum condition emerge from the energy balance; neither is imposed in the code.

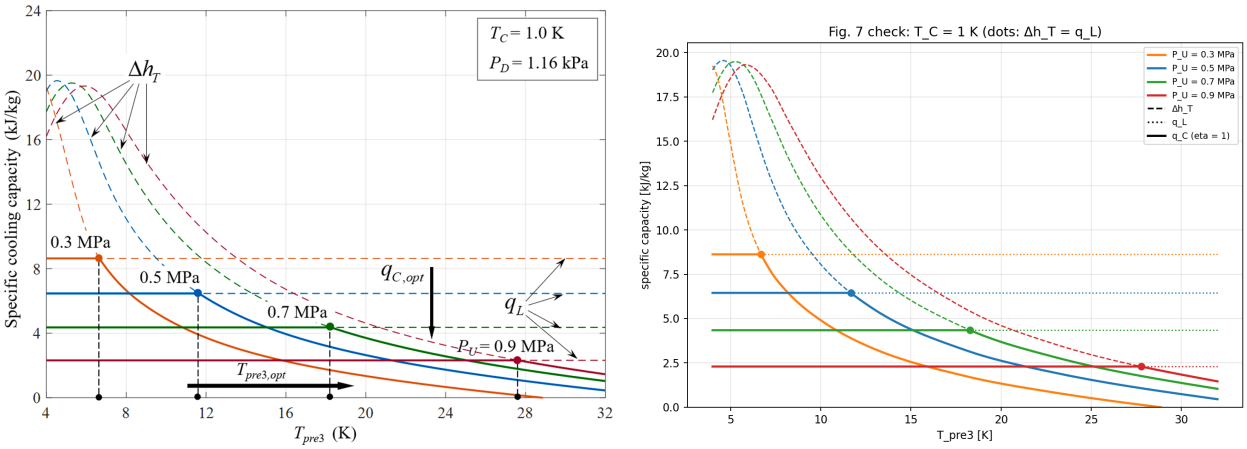


(a) Zhang & Dang (2021)

(b) This work, identical inputs

Figure 20.1: Refrigeration per unit mass q_C versus P_U at $T_{\mathrm{pre3}} = 8$ K (paper Fig. 5). The open circle marks the optimum.

Correlations of the optimum (paper Eqs. 35–36). Over the paper's range of T_{pre3} the solver's $P_{U,\mathrm{opt}}$ differs from Eq. (36) by 1.39 % on average and 3.72 % at worst, and $q_{C,\mathrm{opt}}$ from Eqs. (35)–(36) by 3.12 % on average (largest deviation 10.4 % at 30 K, where q_C is small). Evaluated at the solver's own $P_{U,\mathrm{opt}}$, the paper's q_C correlation agrees to 0.17 % on average (0.47 % worst). For the optimum precooling temperature versus P_U the differences from Eq. (36) at 0.3, 0.5, 0.7 and 0.9 MPa are 0.01, 0.23, 0.34 and 0.78 K. No optimum lies on the boundary of the search range.



(a) Zhang & Dang (2021) (b) This work, identical inputs

Figure 20.2: q_C versus T_{pre3} for several P_U (paper Fig. 7).

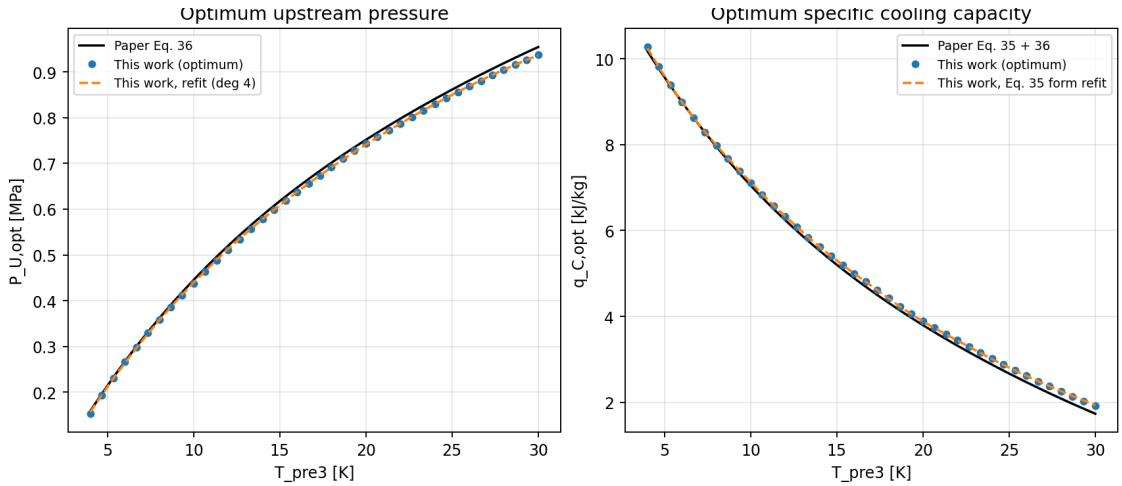


Figure 20.3: Solver optima (markers) against the paper's correlations Eqs. (35)–(36) (lines): $P_{U,\mathrm{opt}}$ and $q_{C,\mathrm{opt}}$ versus T_{pre3}.

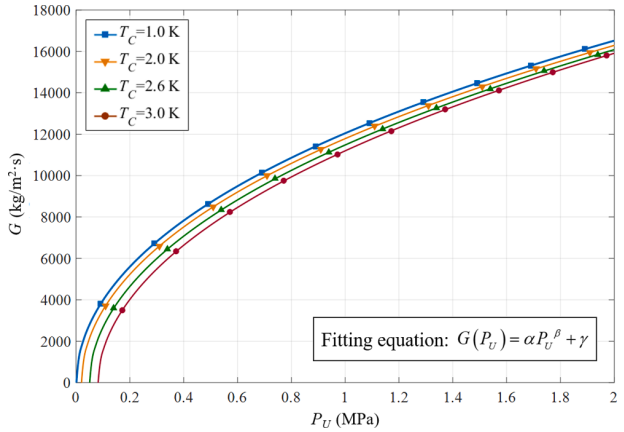
20.2 Mass flux (paper Eqs. 41–49, Table 2, Fig. 11)

T_C [K]	1	2	2.6	3
HEM flux vs. paper Table 2, mean deviation [%]	0.25	0.41	0.51	0.69
HEM flux vs. paper Table 2, worst deviation [%]	0.71	0.72	2.28	1.49
Eq. 49 form refitted to HEM data, worst deviation [%]	0.24	0.53	1.00	1.82

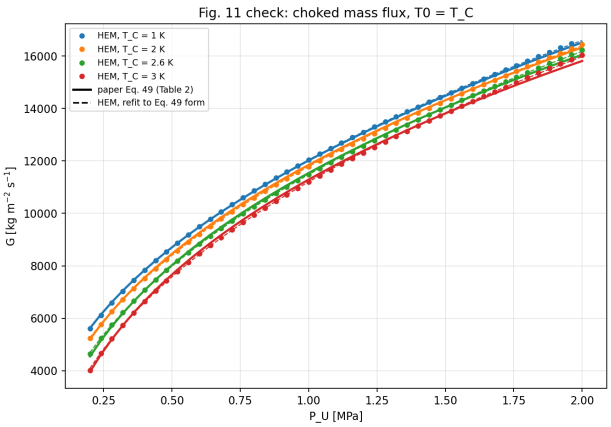
Table 20.1: HEM mass flux against the paper's real-gas flux.

20.3 Gross cooling capacity (paper Fig. 13)

Combining q_C with the HEM flux through $d = 20$ µm gives $Q_C = \dot{m}\, q_C$. With $T_{\mathrm{pre3}} = 8$ K the maximum of Q_C coincides with that of q_C ($P_U = 0.358$ MPa, $Q_C = 18.62$ mW); for a colder last precooler the Q_C maximum moves to a higher pressure than the q_C maximum because $\dot{m}$ increases with P_U (at 5 K: 0.352 vs. 0.213 MPa).

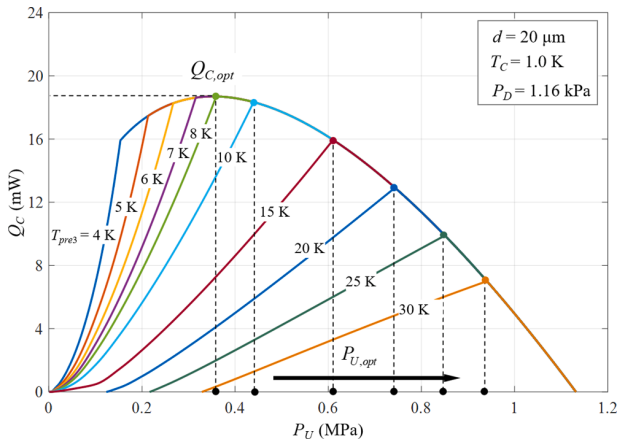


(a) Zhang & Dang (2021)

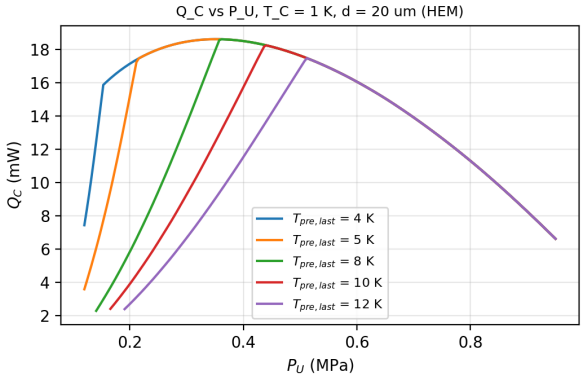


(b) This work, identical inputs

Figure 20.4: Maximum mass flux versus P_U for several T_C (paper Fig. 11).



(a) Zhang & Dang (2021)



(b) This work, identical inputs

Figure 20.5: Gross cooling capacity Q_C versus P_U for several last-stage precooling temperatures (paper Fig. 13).

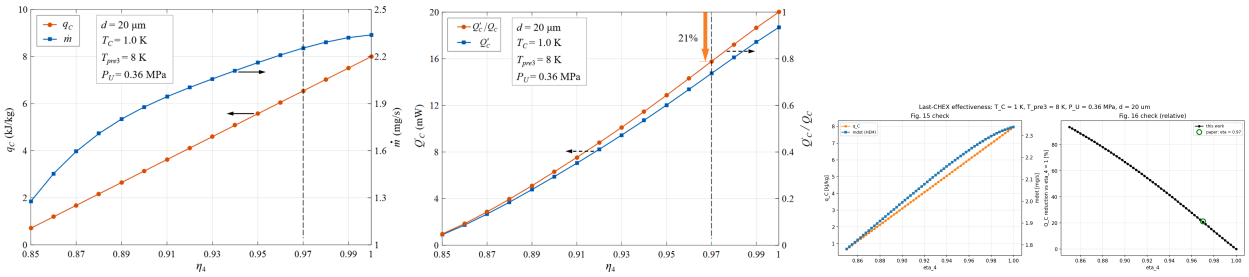
20.4 Effect of the CHEX effectiveness (paper Figs. 14–16)

Keeping all other CHEXs ideal and varying the last effectiveness (optimal $P_U = 0.358$ MPa, $T_{\mathrm{pre3}} = 8$ K, $d = 20$ µm):

η_4	T_8 [K]	T_{11} shortfall [K]	q_C [kJ/kg]	$\dot{m}$ [mg/s]	Q_C [mW]
1	1.000	0	7.961	2.339	18.62
0.995	1.180	0.043	7.718	2.332	18.00
0.99	1.349	0.078	7.475	2.322	17.36
0.98	1.651	0.149	6.989	2.297	16.06
0.97	1.910	0.219	6.503	2.267	14.74
0.95	2.333	0.360	5.531	2.197	12.15
0.90	3.086	0.712	3.102	1.994	6.19

("T_{11} shortfall" is $T_7 - T_{11}$, the temperature by which the return stream leaves the CHEX colder than the warm-end temperature.) Comparison with the paper: $Q_C(\eta = 1) = 18.62$ vs. 18.7 mW, $\dot{m} = 2.339$ vs. 2.34 mg/s, $q_C = 7.96$ vs. 8.0 kJ/kg; $Q_C(\eta = 0.97) = 14.74$ vs. 14.7 mW and a relative drop of 20.8 % vs. about 21 % (paper Fig. 16). The paper's statement that for $\eta_4 \geq 0.97$ the temperature differences at the cold and warm ends should not exceed 1 K and 0.2 K (its Fig. 14) is

met: we find 0.91 K ($T_8 - T_C$) and 0.22 K ($T_7 - T_{11}$) at $\eta = 0.97$. The sensitivities of Q_C are about $-7\,\%$ per 0.01 of η near $\eta = 1$ and the change is almost linear (paper Fig. 16); the contribution of the flux ($\dot{m}$ drops by 3 %) is small compared with that of q_C (18 %) consistent with Eq. (12).



(a) Paper Fig. 15 (b) Paper Fig. 16 (c) This work

Figure 20.6: Effect of the last-CHEX effectiveness η_4 on q_C, $\dot{m}$ and Q_C.

20.5 Precooling loads for the paper's temperatures

At $P_U = 0.36$ MPa and $T_{\text{pre}} = (80, 25, 8)$ K the solver gives $Q_{\text{pre}} = (0.511,\ 9.698,\ 16.613)$ mW and $Q_C = 18.62$ mW, $\dot{m} = 2.339$ mg/s; i.e. the 8 K stage has to absorb about 16.6 mW. Cascade balance Eq. (13) is satisfied to rounding. The solver flags an internal temperature crossover in CHEX 1 (min. gap -0.013 K) and the unchoked flow (the inlet is liquid-like at 8 K, so the HEM maximum is at the back pressure; the flux is still the HEM maximum).

20.6 Summary

The validation comprises 27 checks of three kinds: *anchors* (values read from the paper), *consistency* checks (identities the implementation must satisfy) and *model* checks (HEM against the paper's real-gas flux). All pass; the complete list is given in Appendix D. The largest deviation of a directly reported value is 2.2 %.

Chapter 21

Application to other coolers of the same architecture

To test generality the solver, unchanged, was applied to coolers from two further papers. Only *ideal-model* values reported by those authors are compared, i.e. values computed under the same assumptions (no heat leak, no pressure drop, prescribed effectiveness). Measured performance is deliberately excluded: a measurement contains parasitic loads and line pressure drops that the model omits by construction, so a comparison with it tests the model's assumptions rather than its implementation. The working fluid in both cases is He-4 (CoolProp).

Case	T_C [K]	P_U [MPa]	$\dot{m}$ [mg/s]	T_{pre} [K]; η
Liu et al. (2017)	4.252	1.137	9.1	($60^{\dagger}$, 17.3); ($1^{\dagger}$, $1^{\dagger}$, 0.9857)
Ma et al. (2019), design point	4.5	2.0	10.5	(100, 15); 0.97–1

Table 21.1: Inputs († value not given by the source; it affects only the split of precooling load, not q_C, $\dot{m}$ or Q_C).

Case	Solver	Reported (ideal)	Difference
Liu 2017, Q_C [mW]	53.1	52.5	+1.1 %
Ma 2019 design, Q_C [mW], $\eta = 0.97$–1	90.2–108.8	100	bracketed ($\eta \approx 0.985$)
Ma 2019 design, Q_{pre} at 15 K [mW]	278	260	+7 %

Table 21.2: Solver results against the ideal values reported by the source papers.

Liu et al. (2017), Int. J. Heat Mass Transf. 109, doi:10.1016/j.ijheatmasstransfer.2017.02.070. A He-4 JT stage with supply 1.137 MPa, return 0.104 MPa ($T_C = T_{\mathrm{sat}} = 4.252$ K), $\dot{m} = 9.1$ mg/s and last-stage precooling at 17.3 K; the authors' calculated maximum refrigeration is 52.5 mW. No effectiveness is reported, but the supply enters the last exchanger at 17.3 K and the return leaves it at 17.1 K, so $\eta_3 = 0.9857$ follows from Eq. (16) without fitting. The solver gives $q_C = 5.835$ kJ/kg and $Q_C = 53.1$ mW (+1.1 %). With $\eta_3 = 1$ the result would be about 19 % higher, so the agreement depends on the effectiveness derived from the reported temperatures.

Ma et al. (2019), Cryogenics 104, doi:10.1016/j.cryogenics.2019.102983, design point: 100 mW at 4.5 K, 2.0 MPa, precooling at 100 and 15 K, effectiveness "not lower than 0.97". Over that admissible range the solver gives 90.2 mW ($\eta = 0.97$) to 108.8 mW ($\eta = 1$); the design value corresponds to $\eta \approx 0.985$. The load of the 15 K precooler, 278 mW, is within 7 % of the design value of 260 mW.

Chapter 22

Application: design of a coupled hybrid JT cooler

The validated solver is used the way a designer would use it: choose the target, find the operating point, and read off what the precooling cryocooler has to supply.

22.1 Procedure

1. Create a named case: `python jt_run.py new --fluid he3 --T_C 1.0 --eta 1,1,1,1 --T_pre=-,-,8 --P_U=- --d_um 20`. This produces `Helium_3_1K_20um_4_stage.json`. (Here the last stage temperature is given, 8 K, and the rest is left to the optimiser.)
2. `python jt_run.py feasible cases/Helium_3_1K_20um_4_stage.json`: verdict, $P_{U,\max} = 1.131$ MPa, $T_{\mathrm{pre,opt}}$.
3. `python jt_run.py run cases/Helium_3_1K_20um_4_stage.json`: optimise and solve.

22.2 Result for the paper's cooler

The optimiser finds $P_U = 0.3584$ MPa and $Q_C = 18.62$ mW (both for q_C and Q_C), with $T_{\mathrm{pre,opt}} = 8$ K at that pressure, without being told the optimum; it then selects the upstream temperatures by criterion (15):

PHEX	Gas in [K]	Gas out (T_{pre}) [K]	q [kJ/kg]	Q [mW]
1	26.24	25.64	4.240	9.90
2	12.77	12.29	3.564	8.32
3	8.446	8.00	3.665	8.56
Total precooling			11.47	26.77

The refrigeration $q_C = 7.979$ kJ/kg, $\dot{m} = 2.334$ mg/s (the small difference from 2.339 is the optimal P_U 0.3584 vs. 0.36 MPa). The state table is printed by `run` and stored in `states.csv`; $T_8 = 1$ K, $T_{11} = 8$ K, $h_8 = 5.13$ kJ/kg.

22.3 The upstream map: how to select the precooling cryocooler

Because the upstream temperatures only redistribute a fixed total of 11.47 kJ/kg (26.8 mW), the useful design information is the *map* of loads against temperature. Figure 22.1 sweeps PHEX 1 (left) and PHEX 2 (right) with the others fixed. Reading it:

- Total precooling load is constant (dashed black).
- Raising $T_{\mathrm{pre},1}$ moves load from PHEX 1 to PHEX 2; above ≈ 90 K PHEX 1 carries almost nothing (the exchanger 1 already brings the gas to that temperature) and PHEX 2, the second stage, carries about 18 mW.
- PHEX 2 at 8–25 K trades load with PHEX 3: a colder $T_{\mathrm{pre},2}$ moves load from PHEX 3 to PHEX 2.

The Carnot-weighted cost Φ (Eq. 15) is 263 at the chosen $T_{\mathrm{pre},1} = 26.7$ K (this is the minimum of the sweep; 316 at 12 K, 305–316 above 70 K) and 264 at $T_{\mathrm{pre},2} = 11.1$ K. For a real cryocooler the

capacity curve decides: choose the cooler whose stage capacities at its stage temperatures meet or exceed the loads in this map. For the paper's cooler a four-stage cryocooler with stages near 26, 12 and 8 K carrying about 10, 8 and 9 mW (plus the first stage at 80–300 K carrying no significant load) satisfies the optimum.

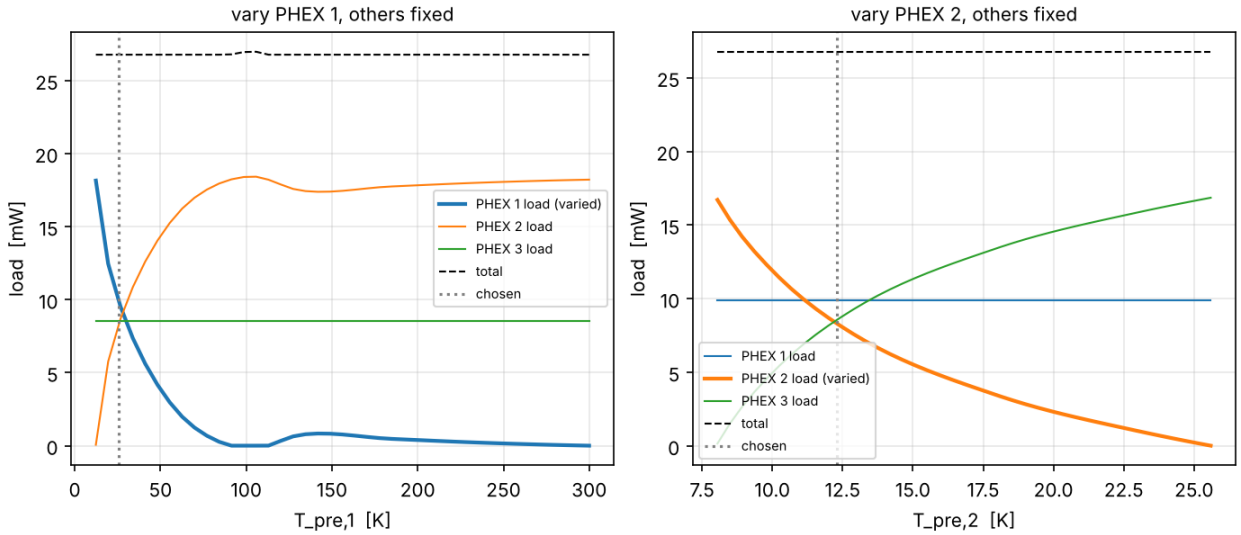


Figure 22.1: Loads of the three PHEXs when PHEX 1 (left) or PHEX 2 (right) temperature is varied; total constant; the dotted line marks the chosen value.

22.4 Comparison with the paper's precooling temperatures

With the paper's choice $T_{\text{pre}} = (80, 25, 8)$ K at $P_U = 0.36$ MPa the loads are $(0.5, 9.7, 16.6)$ mW: the total is the same, but most of it falls on the coldest stage. The Carnot-weighted choice $(25.6, 12.3, 8)$ K spreads it as $(9.9, 8.3, 8.6)$ mW and reduces the work-equivalent cost, since load at 8 K is the most expensive to remove.

22.5 Using the same machinery for another fluid or target

The same three commands work for any supported fluid: e.g. He-4 at 4.5 K, or Ne. The feasibility check will report $P_{U,\max}$, the inversion limit and $T_{\text{pre,opt}}$, so one can see directly whether a target is possible and how warm the last-stage precooler can be, before any optimisation. The cases of Section 21 can be re-run with `--optimize` to obtain the best T_{pre} and P_U for those coolers.

Chapter 23

Limitations

- **Ideal model.** No heat leak, no pressure drop, no axial conduction, discharge coefficient $C_d = 1$ (A2, A3, A9). Results are upper bounds for a real cooler, and the solver is therefore compared only with ideal-model values.
- **Effectiveness is an input.** It is not predicted from exchanger geometry; Q_C falls by about 7 % per 0.01 of η_{last} near $\eta = 1$, so η must come from measurement or a separate exchanger model.
- **Precooling is imposed.** Cryocooler capacity curves are not modelled; the load map states the capacities required.
- **Mass flow** is the HEM maximum through a single ideal orifice, or a prescribed value; compressor coupling and multi-orifice elements are not modelled.
- **He-3 property range:** $T_C \leq 3.2$ K, no extrapolation, a near-critical gap at 101.7–116.4 kPa.
- **Other fluids:** P_U must exceed the critical pressure; accuracy is that of the CoolProp equations of state.
- **Internal pinch** is detected and reported, not corrected.
- **Upstream criterion.** Carnot weighting is one reasonable choice; the actual cryocooler capacity curve should decide.

Chapter 24

Reproducibility

1. `pip install numpy scipy pandas matplotlib CoolProp`; place the He-3 tables in `data/he3/`.
2. `python jt_run.py validate --dense`: paper reproduction, all figures and the 27-check scorecard (grid sizes can be set individually with `--n-curve`, `--n-tpre`, `--n-opt-grid`, `--n-flux`, `--eta-step`).
3. `python make_extra_figures.py --n-pu 200 --outdir results/extra`: Q_C versus P_U curves and He-3 c_p (with progress display).
4. `python run_all.py`: data check, paper reproduction and the other-cooler cases.
5. Design: `python jt_run.py new --fluid he3 --T_C 1.0 --eta 1,1,1,1 --T_pre=-,-,8 --P_U=- --d_um 20`, then `feasible` and `run` on the created case file.

Appendix A

State labelling for general N

For N CHEXs the hot-stream states are numbered $1, \ldots, 2N$ in order along the supply path (CHEX i hot inlet $2i-1$, hot outlet $2i$; the following PHEX leaves state $2i+1$), orifice outlet $2N+1$, evaporator outlet $2N+2$, then the return-stream states $2N+3, \ldots, 4N+2$ from the cold end of CHEX N to the warm exit. For $N = 4$: 1–8 hot, 9 orifice outlet, 10 evaporator outlet, 11–14 return (the paper's numbering, its Fig. 2). The `states.csv` file holds T, P, h, phase and a role string for each.

Appendix B

Flags

Flag	Meaning
`INTERNAL_TEMPERATURE_CROSSOVER`	stream temperatures cross inside a CHEX (warning)
`UNCHOKED_RESTRICTION_FLOW`	HEM maximum flux at the back pressure; flux still the HEM maximum (warning)
`(error class)`	a PHEX would have to heat the gas; balance fails; no net refrigeration

Appendix C

Example case file

```
{ "name": "Liu 2017 He-4 open-cycle JT, last stage at 17.3 K",
  "fluid": "Helium", "T_C": 4.2517, "P_U": 1.137, "T_H": 300,
  "T_pre": [60.0, 17.3], "eta": [1.0, 1.0, 0.9857],
  "mdot_mg_s": 9.1,
  "reference": {"citation": "...", "location": "Sec. 2.2 / Eq. (1)",
                "kind": "...", "note": "..."},
  "reported": {"Q_C_mW": 52.5, "P_D_kPa": 104.0} }
```

Keys beginning with `_` are ignored; `d_um` and `mdot_mg_s` are mutually exclusive.

Appendix D

Scorecard of the paper reproduction

Check	Solver	Paper	Tolerance
$P_{U,\mathrm{opt}}$ at $T_{\mathrm{pre3}} = 8$ K [MPa]	0.3584	0.36	± 0.03
$q_{C,\mathrm{opt}}$ [kJ/kg]	7.979	8.0	± 0.4
$P_{U,\mathrm{max}}$ [MPa]	1.131	1.13	± 0.05
$q_C = \min(\Delta h_T, q_L)$	4×10^{-14}	—	$\leq 10^{-6}$
$\Delta h_T = q_L$ at optimum (Eq. 34)	7.5×10^{-8}	—	≤ 0.01
$P_{U,\mathrm{opt}}$ vs. Eq. 36, mean / worst [%]	1.39 / 3.72	—	5 / 10
$q_{C,\mathrm{opt}}$ vs. Eqs. 35+36, mean [%]	3.12	—	5
$T_{\mathrm{pre3,opt}}$ at 0.3/0.5/0.7/0.9 MPa [K]	0.01/0.23/0.34/0.78	Eq. 36	± 1.5
HEM flux at $T_C = 1, 2, 2.6, 3$ K, mean [%]	0.25/0.41/0.51/0.69	Table 2	15
Eq. 49 form, worst [%]	0.24/0.53/1.00/1.82	—	2
q_C ($\eta = 1$) [kJ/kg]	7.961	8.0	± 0.4
Q_C ($\eta = 1$) [mW]	18.62	18.7	± 1
$\dot{m}$ ($\eta = 1$) [mg/s]	2.339	2.34	± 0.15
Q_C ($\eta = 0.97$) [mW]	14.74	14.7	± 1
Q_C drop at $\eta = 0.97$ [%]	20.82	21.39	± 3

The table groups related checks; there are 27 in total, all of which pass.

References

T. Zhang and H. Dang, "Investigations on a 1 K hybrid cryocooler composed of a four-stage Stirling-type pulse tube cryocooler and a Joule–Thomson cooler. Part A: Theoretical analyses and modeling", *Cryogenics* **116** (2021) 103282. Liu et al., *Int. J. Heat Mass Transf.* **109** (2017) 1094–1099. Ma et al., *Cryogenics* **104** (2019) 102983.